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INTELLIGENT SAMPLING AND INFERENCE FOR MULTIPLE CHANGE POINTS IN EXTREMELY LONG DATA SEQUENCES

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Change point estimation in its offline version is traditionally performed by optimizing a data fit criterion over the data set of interest that considers each data point as the true location parameter. The data point that minimizes the criterion is declared as the change point estimate. For estimating multiple change points, the available procedures in the literature are analogous in spirit, but significantly more involved in execution. Since change-points are local discontinuities, only data points close to the actual change point provide useful information for estimation, while data points far away are superfluous, to the point where using only the points close to the true parameter is just as precise as using the full data set. Leveraging this "locality principle", we introduce a two-stage procedure for the problem at hand: the 1st stage uses a *sparse subsample* to obtain pilot estimates of the underlying change points, and the 2nd stage updates these estimates by sampling densely in appropriately defined neighborhoods around them, and computing a local change-point estimator. We establish that this method achieves the same rate of convergence and even virtually the same asymptotic distribution as an analysis of the full data, while reducing computational complexity to $\sqrt{N}$ time (N being the length of data set) under favorable circumstances, as opposed to at least $O(N)$ time for all current procedure. This makes our new method promising for the analysis of exceedingly long data sets with adequately spaced out change points. The main results, which, in particular, lead to a prescription for constructing explicitly computable joint (asymptotic) confidence intervals for a *growing number* of change-points via the proposed procedure, are established under a signal plus noise model with independent and identically distributed error terms, but extensions to dependent data settings, as well as multiple stage (> 2) procedures are also provided. The performance of our procedure – which is coined "intelligent sampling" – is illustrated on both emulated data and a real internet data stream.

1. Introduction. Change point analysis has been extensively studied in both the statistics and econometrics literature [4], due to its wide applicability in diverse fields, including economics and finance [8], quality control [19], neuroscience [13], etc. A change point represents a discontinuity in the parameters of the data generating process. The literature has investigated both the *offline* and *online* versions of the problem [4, 6]. In the former case, one is given a sequence of observations and questions of interest include: (i) whether there exists a change point and (ii) if there exists one

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(or multiple) change point(s), identify its (their) location, as well as estimate the parameters of the data generating process to the left and right of it (them). In the latter case, one obtains new observations in a sequential manner and the main interest is in quickest detection of the change point. This paper deals with the *offline* problem for very large data sequences.

The offline analysis of *extremely long* data sequences poses a body of challenges, not least owing to the fact that conventional modes of analysis based on the entire data are computationally challenging, owing to the massive size involved. The identification of change-points in a sequence, which constitute local discontinuities, requires some sort of a search procedure. A list of such methods, along with summaries of their results, can be found in [16]. Time complexities for identification of multiple change points range between $O(N)$ and $O(N^2)$ depending on spacing conditions between change points, where N is the length of the time series:[1]such time complexities are not attractive for very long sequences.

Such offline scenarios arise when network engineers examine *previously stored traces of traffic* in high-speed computer and communication networks and examine them in greater detail aiming to identify shifting persistent patterns followed by attribution analysis, possibly correlating the detected change points with other data that would point to either infrastructure related issues or shifts in user patterns. This should be contrasted with *online monitoring* where the goal is to raise alarms *in real time* whenever the underlying process is "out-of-control" and notify network engineers, who determine the tolerance level for alarms, i.e. how large the change should be to be considered negatively impacting network operations [21], [1]. In the *offline setting*, the number of observations in network traces, corresponding to the number of packets or their payload in bytes at a granular temporal scale, is in the thousands per minute. Analogous problems come up in other systems including manufacturing processes [20], or other cyber-physical systems equipped with sensors [11], where data may possibly be stored in a distributed manner.

The point of view adopted in the paper is as follows: we are given a data sequence of massive length and are interested in identifying the major changes in this series. While *short-lived* changes of intensities in mean shifts may occur frequently, such transient perturbations do not affect the overall performance of most engineered or physical systems. With such applications in mind, it is very reasonable to assume that the number of truly significant changes that *persist* over time, is *not particularly large*. To communicate the core ideas effectively, we employ a canonical model: a long piece-wise constant mean model with multiple jumps – where the number of jumps increases at a rate much slower than the length of the series — that are not too small relative to the fluctuations induced by noise. Also, the flat stretches between consecutive jumps are taken to be persistent. Indeed, the 'piece-wise constant with jumps' strategy has been considered by a variety of authors (see [16] and references therein) for the problem at hand and is convenient for developing the methodology. We demonstrate subsequently that our proposed strategy is robust to mis-specifications of the persistent piecewise constant model, in the sense that its detection of persistent stretches is essentially unaffected in the presence of short-lived 'idiosyncratic' signal bursts.

[1]Precise identification of change-points under very relaxed spacing conditions can be accomplished, for example, by a dynamic programming based algorithm, which is of order N^2.

Next, we summarize key contributions of this paper.

1. We propose an effective solution to the computational problem discussed above via a strategy called "intelligent sampling", which proceeds by making two (or more) passes over the time-series at different levels of resolution. The first pass is carried out at a coarse time resolution and enables the analyst to obtain pilot estimates of the change-points, while the second investigates stretches of the time-series in (relatively) small neighborhoods of the initial estimates to produce updated estimates enjoying a high level of precision; in fact, essentially the same level of precision, as would be achieved by an analysis of the entire massive time-series. The core advantage of our proposed method is that it reduces computational time from *linear* to *sub-linear* under appropriate conditions, and, in fact close to square-root order, if the number of change-points is small relative to the length of the time-series. It is established that the computational gains (and analogously other processing gains from input-output operations) can be achieved *without compromising* any statistical efficiency. From a realistic stand-point, the time resolution to be used for the first pass is difficult to determine beforehand, and to alleviate this methodological problem we propose an adaptive *doubling algorithm* [Section 5.4] that samples at increasing levels of resolution till a sufficient number of initial change point estimates are found. As the emulation experiment in Section 6 demonstrates, our proposed method consistently picks up persistent structural changes in the presence of multiple spiky signals while staying agnostic to the latter. Nevertheless, if short duration shifts are also of interest, the interested researcher can further analyze the persistent segments (which are of smaller order than N) with any of the available procedures in the literature in a *parallel* fashion, thus retaining the computational gains achieved by intelligent sampling.

2. Most of our results are rigorously developed in the signal plus noise model with independent and identically distributed (iid) sub-Gaussian errors, and in certain cases, normal errors. Normal errors have been widely used in the change point literature to showcase methods and establish theoretically their performance guarantees, e.g. [9, 17, 24], since they provide an attractive canonical model and are more amenable to analysis. However, we also provide results and indicate extensions when the errors exhibit short or long-range dependence and also in the presence of non-stationarity, since both these features are likely to be present with very long data sequences. Some empirical evidence is provided to this effect.

Further, the focus of the presentation is on 2-stage procedures that provide all the key insights into the workings of the intelligent sampling procedure. However, for settings where the size of the data exceeds 10^{10}, multiple stages are required to bring down the analyzed subsample to a manageable size. We therefore cover extensions to multi-stage intelligent sampling procedures as well as address how samples should be allocated at these different stages. Furthermore, such massive data sequences, often, can not be effectively stored in a single location. This does not pose a problem for intelligent sampling as it adapts well to distributed computing: it can be applied on the reduced size subsamples at the various locations where the original data are stored, followed by a single subsequent back and forth communication between the various locations and the central server, and subsequent calculations essentially carried out on the local servers. This is elaborated on in Section 9.3.

3. On the inferential front, we establish asymptotic approximations to the joint distribution of the change-point estimates obtained by intelligent sampling in terms of the distribution functions of

(typically asymmetrically) drifting random walks on the set of integers [Theorems 5 and 4], which can then be used to provide explicitly computable asymptotic joint confidence intervals for both finitely many and a *growing number* of change points. While concentration properties of change-point estimates around the the true parameters in multiple change point problems are known, to the best of our knowledge, such results involve hard-to-pin-down constants (this is discussed in some more detail in Section 9.1 in the context of binary segmentation) and are therefore difficult to use in practical settings. Our prescribed methods involve estimating signal to noise ratios at the different change-points, which is easy to accomplish, and values of quantiles of drifting Gaussian random walks for different values of the drift parameter (which can be pre-generated on a computer). The arguments involved in establishing Theorem 5 require, among other things, careful analyses of the distribution functions of both symmetric and asymmetric Gaussian random walks (see part B of Supplement, Section 9) which may very well prove useful in many other contexts. To the best of our knowledge, our prescription of joint asymptotic confidence intervals for a growing number of change-points with readily computable estimates of the distributional parameters involved [see Theorem 5] is the *first of its kind* in the change point literature.

4: We illustrate our procedure by applying it to two data-sets, the first in Section 6.1 on a partly emulated data set where the error process arises from real network traffic data and we create ground-truths by 'injecting' change-points into the error process through the addition of piecewise constant mean shifts. Section 6.2 examines a real data set, where our intelligent sampling proposal is useful. It corresponds to network (internet) traffic destined for an autonomous system - a collection of Internet addresses under the control of a single network operator. The corresponding time series reflects aggregate incoming traffic to the autonomous system at 2 sec resolution. Short lived spikes are of less interest to network operators, since they usually correspond to transient traffic patterns. On the other hand, persisting shifts are of significant interest, indicating possible malicious activity, technical issues with the infrastructure, or at larger time scales, traffic growth that needs to be mitigated by corresponding capacity growth of the network's infrastructure. While this data is better described by a piecewise linear regime (as opposed to the piecewise constant model used for our theoretical development), our intelligent sampling technique continues to work well after modifying it to fit piecewise linear models at each stage: the procedures to obtain the first stage pilot estimates, as well as the second stage updated estimates are simply tweaked so that they can tackle piecewise linear functions. This demonstrates the scope of intelligent sampling to data-sets with more general mean structures than the one tackled in depth in this paper, and makes it attractive as a general principle.

The remainder of the paper is organized as follows. Section 2 addresses intelligent sampling for the simpler *single* change point problem, which provides some fundamental insights into the nature of the procedure and its theoretical and computational properties. Section 3 deals with the main topic of this study: intelligent sampling for the multiple change point problem with a growing number of change-points, and presents the main theoretical results of the paper. Section 4 develops the practical methodology for intelligent sampling using *binary segmentation* as the working procedure at Stage 1, and studies the computational complexity of the resulting approach. Section 9.3 provides an elaborate study of the minimum subsample size required for precise inferences as a function of the length of the full data sequence and the signal-to-noise ratio using multiple stage procedures.

Extensions to non-iid settings which are more pertinent for the more special case of time-series data are discussed in Section 9.4. The numerical performance of the procedure is calibrated via thorough simulations in Section 5, while applications to both emulated and real Internet data are presented in 6. Section 7 concludes with a discussion of possible extensions of the intelligent sampling procedure, both in terms of alternative Stage 1 procedures (like wild binary segmentation and SMUCE), and also to other kinds of data (non-Gaussian data, discrete data, decaying signals). In the interests of space, all proofs and elaborate discussions of various facets of intelligent sampling are collected in the Supplement, which is made publicly available on the online repository at `https://github.com/jlnlu/Intelligentsamplingdraft`.

2. Intelligent Sampling for the Single Change Point Problem.

2.1. *Single Change Point Model.* The simplest possible setting for the change point problem is the *stump* model, where data $(1/N, Y_1), \cdots, (N/N, Y_N)$ are available with $Y_i = f(i/N) + \varepsilon_i$ for $i = 1, \cdots, N$, and where the error term ε_i is independent and identically distributed (iid) following a $N(0, \sigma^2)$ distribution, while the function f takes the form

$$f(x) = \alpha \cdot 1(x \leq \tau) + \beta \cdot 1(x > \tau), \qquad x \in (0, 1), \tag{1}$$

for some constants $\alpha, \beta \in \mathbb{R}$, $\alpha \neq \beta$, and $\tau \in (0, 1)$: the so-called 'stump' model. For estimating the *change point* τ we employ a least squares criterion, given by

$$(\hat{\alpha}, \hat{\beta}, \hat{\tau}) := \underset{(a,b,t)\in\mathbb{R}^2\times(0,1)}{\arg\min} \sum_{i=1}^{N} (Y_i - a \cdot 1(i/N \leq t) - b \cdot 1(i/N > t))^2 . \tag{2}$$

Using techniques similar to those in Section 14.5 of [14], we can establish that the estimator $\hat{\tau}$ is consistent for $\tau_N := \lfloor N\tau \rfloor / N$, which acts as the change point among the covariates lying on the even grid.

PROPOSITION 1. *For the stump model with normal errors the following hold:*
(i) Both $(\hat{\alpha} - \alpha)$ and $(\hat{\beta} - \beta)$ converge to 0 with rate $O_p(N^{-1/2})$.
(ii) The change point estimate $\hat{\tau}$ satisfies

$$\mathbb{P}\left[N(\hat{\tau} - \tau_N) = k\right] \to \mathbb{P}[L = k] \text{ for all } k \in \mathbb{Z} \tag{3}$$

where $L = \underset{i\in\mathbb{Z}}{\arg\min}\, X(i)$, and the random walk $\{X(i)\}_{i\in\mathbb{Z}}$ is defined as

$$X(i) = \begin{cases} \Delta(\varepsilon_1^* + \ldots + \varepsilon_{i-1}^* + \varepsilon_i^*) + i\Delta^2/2, & i > 0 \\ 0, & i = 0 \\ -\Delta(\varepsilon_{i+1}^* + \ldots + \varepsilon_{-1}^* + \varepsilon_0^*) + |i|\Delta^2/2, & i < 0, \end{cases} \tag{4}$$

with $\varepsilon_0^*, \varepsilon_1^*, \varepsilon_2^*, \ldots$ and $\varepsilon_{-1}^*, \varepsilon_{-2}^*, \ldots$ being iid $N(0, \sigma^2)$ random variables and $\Delta := \beta - \alpha$.

Next, we make several notes on the random variable L introduced in the above proposition, as it appears multiple times throughout the remainder of this paper. Although, at a glance, the distribution of L depends on two parameters, Δ and σ, in actuality L is completely determined by the signal-to-noise ratio Δ/σ due to the Gaussian setting. To see this, note that we can re-write $L = \arg\min_{i\in\mathbb{Z}}(Z_i/|\Delta\sigma|)$ where

$$\frac{X(i)}{|\Delta\sigma|} = \begin{cases} \text{sgn}(\Delta)(\varepsilon_1^*/\sigma + \cdots + \varepsilon_i^*/\sigma) + i|\Delta/\sigma|/2, & i > 0 \\ 0, & i = 0 \\ -\text{sgn}(\Delta)(\varepsilon_{i+1}^*/\sigma + \cdots + \varepsilon_0^*/\sigma) + |i\Delta/\sigma|/2, & i < 0 \end{cases} \tag{5}$$

Since $\{\text{sgn}(\Delta)\varepsilon_i^*/\sigma\}_{i\in\mathbb{Z}}$ are iid $N(0,1)$ random variables, invariant under Δ and σ, it follows that L only depends on the single parameter Δ/σ. Hence, from here on, denote the associated random process as

$$X_\Delta(i) = \begin{cases} \text{sgn}(\Delta)(\varepsilon_1^\diamond + \cdots + \varepsilon_i^\diamond) + i|\Delta|/2, & i > 0 \\ 0, & i = 0 \\ -\text{sgn}(\Delta)(\varepsilon_{i+1}^\diamond + \cdots + \varepsilon_0^\diamond) + |i| \cdot |\Delta|/2, & i < 0 \end{cases} \tag{6}$$

where $\varepsilon_j^\diamond$ for $j \in \mathbb{Z}$ are all iid $N(0,1)$ random variables. Denote the argmin of the random walk $X_\Delta(i)$ as $L_\Delta = \arg\min_{i\in\mathbb{Z}} X_\Delta(i)$. An immediate observable property of L_Δ is the stochastic ordering with respect to $|\Delta|$:

PROPOSITION 2. *Suppose we have constants $\Delta_1, \Delta_2 \in \mathbb{R}$ such that $0 < |\Delta_1| < |\Delta_2|$, then for any positive integer k*

$$\mathbb{P}[|L_{\Delta_1}| \le k] \le \mathbb{P}[|L_{\Delta_2}| \le k] \tag{7}$$

Practically, this stochastic ordering implies that if the $1-\alpha$ quantile $Q_{\Delta_1}(1-\alpha)$ of $|L_{\Delta_1}|$ is known, then $Q_{\Delta_1}(1-\alpha)$ can also serve as a conservative $1-\alpha$ quantile of $|L_{\Delta_2}|$ for any $|\Delta_2| \ge |\Delta_1|$. This can be useful in settings where given $J > 0$ random variables L_{Δ_i} for $i = 1\ldots, J$, we desire positive integers ℓ_i for $i = 1, \ldots, J$ such that $\mathbb{P}[|L_{\Delta_i}| \le \ell_i] \ge 1-\alpha$ for $i = 1, \ldots, J$. This scenario will appear in later sections where we consider models containing several change points with possibly different jump sizes. In such situations, a simple solution is to take $\ell_i = Q_{\Delta_m}(1-\alpha)$ for all i where $m = \arg\min_{1,\ldots,J}|\Delta_i|$, or in other words letting each ℓ_i be the $1-\alpha$ quantile of the $|L_{\Delta_i}|$ with the smallest parameter. Alternatively we can generate a table of quantiles for distributions $L_{\delta_1}, L_{\delta_2}, L_{\delta_3}, \ldots$ for a mesh of positive constants $\delta_1 < \delta_2 < \ldots$ (e.g. we can let the $\delta_j = 0.1j$ for $j = 5, \ldots, 1000$), and let $\ell_i = Q_{\delta_j}(1-\alpha)$ where $\delta_j = \max\{\delta_k : \delta_k \le \Delta_i\}$, for $i = 1, \ldots, J$.

2.2. *The Intelligent Sampling Procedure and its Properties.*

(ISS1): From the full data set of $\left(\frac{1}{N}, Y_1\right), \left(\frac{2}{N}, Y_2\right), ..., (1, Y_N)$, take an evenly spaced subsample of approximately size $N_1 = K_1 N^\gamma$ for some $\gamma \in (0,1)$, $K_1 > 0$: thus, the data points are $\left(\frac{\lfloor N/N_1\rfloor}{N}, Y_{\lfloor N/N_1\rfloor}\right)$, $\left(\frac{2\lfloor N/N_1\rfloor}{N}, Y_{2\lfloor N/N_1\rfloor}\right)$, $\left(\frac{3\lfloor N/N_1\rfloor}{N}, Y_{3\lfloor N/N_1\rfloor}\right)$ ...

(ISS2): On this subsample apply least squares to obtain estimates $\left(\hat{\alpha}^{(1)}, \hat{\beta}^{(1)}, \hat{\tau}_N^{(1)}\right)$ for parameters (α, β, τ_N).

By the results for the single change-point problem presented above, $\hat{\tau}_N^{(1)} - \tau_N$ is $O_p(N^{-\gamma})$. Therefore, if we take $w(N) = K_2 N^{-\gamma+\delta}$ for some small $\delta > 0$ (much smaller than γ) and any constant $K_2 > 0$, with probability increasing to 1, $\tau_N \in [\hat{\tau}_N^{(1)} - w(N), \hat{\tau}_N^{(1)} + w(N)]$. In other words, this provides a neighborhood around the true change point as desired; hence, in the next stage only points within this interval will be used.

(ISS3): Fix a small constant $\delta > 0$. Consider all i/N such that $i/N \in [\hat{\tau}_N^{(1)} - K_2 N^{-\gamma+\delta}, \hat{\tau}_N^{(1)} + K_2 N^{-\gamma+\delta}]$ and $(i/N, Y_i)$ was not used in the first subsample. Denote the set of all such points as $S^{(2)}$.

(ISS4): Fit a step function on this second subsample by minimizing

$$\sum_{i/N \in S^{(2)}} \left(Y_i - \hat{\alpha}^{(1)} 1(i/N \leq d) - \hat{\beta}^{(1)} 1(i/N > d)\right)^2$$

with respect to d, and take the minimizing d to be the second stage change point estimate $\hat{\tau}_N^{(2)}$.

The next theorem establishes that the intelligent sampling estimator $\hat{\tau}_N^{(2)}$ is *consistent with the same rate of convergence* as the *estimator based on the full data*.

THEOREM 1. *For the stump single change point model, the estimator obtained based on intelligent sampling satisfies*

$$|\hat{\tau}_N^{(2)} - \tau_N| = O_p(1/N).$$

PROOF. See Section 8.2.1 in Supplement Part A, where a result for a more general model is proven. □

To derive a clean statement of the asymptotic distribution, we introduce a slight modification to the definition of the true change point and define a new type of 'distance' function $\lambda_2 : [0,1]^2 \mapsto \mathbb{Z}$, as follows. First, for convenience, denote the set of i/N's of the first stage subsample as

$$S^{(1)} := \left\{ \frac{i}{N} : i \in \mathbb{N},\, i < N,\, i \text{ is divisible by } \lfloor N/N_1 \rfloor \right\}, \tag{8}$$

then for any $a, b \in (0,1)$

$$\lambda_2(a,b) := \begin{cases} \sum_{i=1}^{N} 1\left(a < \frac{i}{N} \leq b,\, \frac{i}{N} \notin S^{(1)}\right) & \text{if } a \leq b\,, \\ -\sum_{i=1}^{N} 1\left(b < \frac{i}{N} \leq a,\, \frac{i}{N} \notin S^{(1)}\right) & \text{otherwise.} \end{cases} \tag{9}$$

The modified "distance" $\hat{\tau}_N$ is $\lambda_2(\tau_N, \hat{\tau}_N^{(2)})$, instead of $N(\hat{\tau}_N^{(2)} - \tau_N)$, does converge weakly to a distribution as the next results establishes.

THEOREM 2. *For any integer ℓ,*

$$\mathbb{P}\left[\lambda_2\left(\tau_N, \hat{\tau}_N^{(2)}\right) = \ell\right] \quad \to \quad \mathbb{P}[L_{\Delta/\sigma} = \ell]\,. \tag{10}$$

PROOF. See Section 8.2.2 in Supplement Part A where this is established for a more general model. □

Computational gains: The results above establish that the two stage procedure can, using a subset of the full data, be asymptotically almost as precise as employing the full data set. In practice this allows for quicker estimation of big data sets without losing precision. The first stage uses about $N_1 \sim N^\gamma$ points to perform least squares fitting of a stump model, and this step takes $O(N^\gamma)$ computational time. The second stage applies a least-squares fit of a step function on the set $S^{(2)}$, which contains $O(N^{1-\gamma+\delta})$ points and therefore uses $O(N^{1-\gamma+\delta})$ time.

Hence, the two stage procedure requires order $N^\gamma \vee N^{1-\gamma+\delta}$ computation time, which is minimized by setting $\gamma = 1-\gamma+\delta$, or $\gamma = \frac{1+\delta}{2}$. As δ tends to 0 (any small positive value of δ yields the above asymptotic results), the optimal γ tends to 1/2. Therefore, one should employ $N_1 = \sqrt{N}$ at the first stage and the second stage sample should be all points in the interval $[\hat{\tau}_N^{(1)} - K_2\sqrt{N}, \hat{\tau}_N^{(1)} + K_2\sqrt{N}]$, minus those at the first stage, where K_2 ensures that this interval contains τ with an acceptable high probability $1-\alpha$. If one knows the jump size Δ, K_2 can be determined as the $1-\frac{\alpha}{2}$ quantile of the random variable $L_{\Delta/\sigma}$; in the realistic unknown Δ case, a lower estimate of Δ can yield a corresponding conservative value of K_2.

REMARK 1. *The λ_2 distance was introduced above because it is generally not possible to derive an asymptotic distribution for $N(\hat{\tau}_N^{(2)} - \tau_N)$. Indeed, one can manufacture parameter settings quite easily, that produce different limit distributions along different subsequences. For a specific example, see Remark 10 in Supplement Part A.*

REMARK 2. *The 2-stage procedure can be extended to multiple stages. In the 2-stage version, we first use a subsample of size N^γ to find some interval $[\hat{\tau}^{(1)} - K_1 N^{-\gamma+\delta_1}, \hat{\tau}^{(1)} + K_1 N^{-\gamma+\delta_1}]$ which contains the true value of τ with probability going to 1. However, it is possible to refrain from using all the data points in the interval at the second stage. Instead, a 2-stage procedure can be employed as follows: take a subset of N^ζ (for some $0 < \zeta < 1-\gamma+\delta_1$) points from the second stage interval and obtain an estimate $\hat{\tau}^{(2)}$ and an interval $[\hat{\tau}^{(2)} - K_2 N^{-\gamma-\zeta+\delta_1+\delta_2}, \hat{\tau}^{(2)} + K_2 N^{-\gamma-\zeta+\delta_1+\delta_2}]$ (note that δ_1 and δ_2 can be as small as one pleases) which contains τ with probability going to 1. In the third stage, all points in the aforementioned interval (leaving aside those used in previous stages) are used to obtain the final estimate $\hat{\tau}^{(3)}$.*

Such a procedure will have the same rate of convergence as the one using the full data: $(\hat{\tau}^{(3)} - \tau) = O_p(1/N)$, and the same asymptotic distribution (in terms of a "third stage distance" similar to how λ_2 was defined) as the one and two stage procedures. In terms of computational time, the first stage takes $O(N^\gamma)$ time, the second stage $O(N^\zeta)$ time, and the final stage $O(N^{1-\gamma-\zeta+\delta_1+\delta_2})$ time, for a total of $O((N^\gamma \vee N^\zeta \vee N^{1-\gamma-\zeta+\delta_1+\delta_2}))$ time, which can reach almost $O(N^{1/3})$ time. In

general, a K stage procedure, which works along the same lines can operate in almost as low as $O(N^{1/K})$ *time.*

3. The Case of Multiple Change Points. Suppose one has access to a data set $Y_1, Y_2, \ldots, Y_N$ generated according to the following model:

$$Y_i = \theta_i + \varepsilon_i, \qquad i = 1, 2, 3, ..., N, \tag{11}$$

where the θ_i's form a piecewise constant sequence for any fixed N and the ε_i's are zero-mean error terms[2]. The signal is flat apart from jumps at some unknown change points $1 = \tau_0 < \tau_1 < ... < \tau_J < \tau_{J+1} = N$: i.e. $\theta_{i_1} = \theta_{i_2}$ whenever $i_1, i_2 \in (\tau_j, \tau_{j+1}]$ for some $j \in \{0, ..., J\}$. The number of change points $J = J(N)$ is also unknown and needs to be estimated from the data. We impose the following basic restrictions on this model:

(M1): there exists a constant $\bar{\theta} \in (0, \infty)$ not dependent on N, such that $\max_{i=1,...,N} |\theta_i| \leq \bar{\theta}$;
(M2): there exists a constant $\underline{\Delta}$ not dependent on N, such that $\min_{i=0,...,J} \left|\theta_{\tau_{i+1}} - \theta_{\tau_i}\right| \geq \underline{\Delta}$;
(M3): there exists a $\Xi \in [0, 1)$ and some $C > 0$, such that $\delta_N := \min_{i=0,...,J} (\tau_{i+1} - \tau_i) \geq C N^{1-\Xi}$ for all large N;
(M4): ε_i for $i = 1, ..., N$ are independent centered subgaussian random variables, with subgaussian parameters[3] $\varepsilon_1, \ldots, \varepsilon_N$ that are all bounded above by a constant $\sigma_{\max} > 0$ which is not dependent on N.

REMARK 3. *The third assumption above stipulates that the minimum gap between two consecutive stretches is bounded away from 0. This is areasonable assumption for identifying long and significantly well-separated persistent stretches in a big data setting. Condition (M4) places a restriction on the tail probability of the error terms, but still accommodates for some heteroscedastic behavior (a concern for long data sequences) by allowing noise sequence comprising of independent random variables with different distributions.*

3.1. *Intelligent Sampling on Multiple Change Points.* The intelligent sampling procedure in the multiple change-points case works in two (or more) stages as follows: in the two-stage version, as in Section 2, the first stage aims to find rough estimates of the change points using a uniform subsample (Steps ISM1-ISM4) and the second stage produces the final estimates (Steps ISM5 and ISM6).

(ISM1): Start with a data set $Y_1, \cdots, Y_N$ described in (11).
(ISM2): Take $N_1 = K_1 N^\gamma$ for some K_1 and $\gamma \in (\Xi, 1)$ such that $N/N_1 = o(\delta_N)$; for $j = 1, \cdots, N^*$ where $N^* := \left\lfloor \frac{N}{\lfloor N/N_1 \rfloor} \right\rfloor$, consider the subsample $\{Z_j\} = \{Y_{j\lfloor N/N_1 \rfloor}\}$.

[2]Specifically, we consider the triangular array of sequences $\theta_{i,N}$, which are piecewise constant in i. The error terms $\varepsilon_i = \varepsilon_{i,N}$ also form a triangular array, but we suppress the notation for brevity

[3]The subgaussian parameter of a variable X is any value $\sigma > 0$ such that $\mathbb{E}[\exp(sX)] \leq \exp(s^2\sigma^2/2)$ for all real values s

The subsample $Z_1, Z_2, ...$ can also be considered a data sequence structured as in (11), and since $\delta_N >> N/N_1$, there are jumps in the signal at $\tau_j^* := \left\lfloor \frac{\tau_j}{\lfloor N/N_1 \rfloor} \right\rfloor$ for $j = 1, ..., J$, with corresponding minimum spacing

$$\delta_{N^*}^* := \min_{i=1,...,J+1} |\tau_i^* - \tau_{i-1}^*| = \frac{1}{\lfloor N/N_1 \rfloor} \left(\min_{i=1,\cdots,J+1} |\tau_i - \tau_{i-1}| + O(1) \right) = \left(\frac{N_1}{N} \delta_N \right) (1 + o(1)) . \tag{12}$$

(ISM3): Apply some multiple change point estimation procedure (such as binary segmentation) to the set of Z_i's to obtain estimates $\hat{\tau}_1^*, ..., \hat{\tau}_{\hat{J}}^*$ for the τ_i^*s and $\hat{\nu}_0^{(1)}, \ldots, \hat{\nu}_{\hat{J}}^{(1)}$ for the levels $(\nu_0, \nu_1, \ldots, \nu_J) = (\theta_1, \theta_{\tau_1+1}, \theta_{\tau_2+1}, \ldots, \theta_{\tau_J+1})$.

- the choice of the procedure does not matter so long as the estimates satisfy

$$\mathbb{P}\left[\hat{J} = J, \max_{i=1,...,J} |\hat{\tau}_i^* - \tau_i^*| \le w^*(N^*), \max_{i=0,...,J} |\hat{\nu}_i^{(1)} - \nu_i| \le \rho_N \right] \to 1 \tag{13}$$

for some sequence $w^*(N^*)$ such that $w^*(N^*) \to \infty$, $w^*(N^*) = o(\delta_{N^*}^*)$ and $\rho_N \to 0$.

(ISM4): Convert these into estimates for the τ_i's by letting $\hat{\tau}_j^{(1)} := \hat{\tau}_j^* \lfloor N/N_1 \rfloor$ for $j = 1, ..., \hat{J}$.

- taking $w(N) := (w^*(N^*) + 1)\lfloor N/N_1 \rfloor$, expression (13) gives

$$\mathbb{P}\left[\hat{J} = J, \max_{i=1,...,J} |\hat{\tau}_i^{(1)} - \tau_i| \le w(N), \max_{i=0,...,J} |\hat{\nu}_i^{(1)} - \nu_i| \le \rho_N \right] \to 1. \tag{14}$$

- as a consequence of conditions in (ISM3), $w(N) \to \infty$, $w(N) = o(\delta_N)$, and $w(N) > CN^{1-\gamma}$ for some constant C.

(ISM5): Fix any integer $K > 1$, and consider the intervals $\left[\hat{\tau}_i^{(1)} - Kw(N), \hat{\tau}_i^{(1)} + Kw(N) \right]$ for $i = 1, ..., \hat{J}$. Denote by $S^{(2)}\left(\hat{\tau}_i^{(1)} \right)$ all integers in this interval not divisible by $\lfloor N/N_1 \rfloor$.

(ISM6): For each $i = 1, ..., \hat{J}$, let

$$\hat{\tau}_i^{(2)} = \underset{d \in S^{(2)}\left(\hat{\tau}_i^{(1)}\right)}{\arg\min} \left(\sum_{j \in S^{(2)}\left(\hat{\tau}_i^{(1)}\right)} \left[Y_j - (\hat{\nu}_{i-1}^{(1)} 1(j < d) + \hat{\nu}_i^{(1)} 1(j \ge d)) \right]^2 \right) . \tag{15}$$

The intervals $[\hat{\tau}_j^{(1)} \pm Kw(N)]$ referred to at stage ISM4, all respectively contain $[\tau_j \pm (K-1)w(N)]$ with probability going to 1. These latter intervals have width going to ∞, and both of these intervals contain exactly one change point (as their widths are $O(w(N)) = o(\delta_N)$). Hence, with probability $\to 1$ the multiple change point problem has simplified to $\hat{J}$ single change point problems, justifying ISM6 where a stump model is fitted inside each of $S^{(2)}(\hat{\tau}_j^{(1)})$'s.

REMARK 4. **Multistage Intelligent Sampling in the multiple change-point problem:** *We can also consider intelligent sampling with multiple* (> 2) *stages of estimation for model (11).*

An m-stage intelligent sampling procedure would entail:
a. *Take a uniform subsample* $Y_{\lfloor N/N_1 \rfloor}, Y_{2\lfloor N/N_1 \rfloor}, Y_{3\lfloor N/N_1 \rfloor}, \ldots$, *where* $N_1 = KN^{\gamma}$ *for some* $K > 1, \gamma \in (0,1)$, *to obtain estimates* $\hat{J}$, $\hat{\tau}_1^{(1)}, \ldots, \hat{\tau}_{\hat{J}}^{(1)}$, *and confidence intervals* $\left[\hat{\tau}_j^{(1)} - w(N), \hat{\tau}_j^{(1)} + w(N)\right]$, $1 \leq j \leq \hat{J}$, *for the change points.*
b. *On each interval* $\left[\hat{\tau}_j^{(1)} - w(N), \hat{\tau}_j^{(1)} + w(N)\right]$ *for* $1 \leq j \leq \hat{J}$ *perform the* $(m-1)$ *stage intelligent sampling procedure for the single change point (as described in Remark 2). For narrative purposes, we will focus our analysis mainly towards the two stage procedure.*

Asymptotic behavior of the intelligent sampling based estimators: Next, we present results on the large sample properties of the intelligent sampling estimates. There are several types of results we will showcase to characterize the asymptotic behavior, with the first result focusing on the rate of convergence:

THEOREM 3. *Suppose conditions (M1) to (M4) are satisfied and the first stage estimates satisfy the consistency result (14). Then, for any* $\varepsilon > 0$, *there exist constants* C_1 *and* C_2 *(depending on* ϵ*) such that*

$$\mathbb{P}\left[\hat{J} = J, \max_{k=1,\ldots,J} |\hat{\tau}_k^{(2)} - \tau_k| \leq C_1 \log(J) + C_2\right] \geq 1 - \varepsilon \tag{16}$$

for all sufficiently large N.

PROOF. See Section 9.5 of Appendix B. □

The rate of convergence matches that of certain estimators when applied to the full data (see the convergence properties of wild binary segmentation in Theorem 3.2 of [9]), consistent with the proposal that the two stage estimator is on the same level of accuracy as methods using the full data set.

To further characterize the asymptotic behavior of the intelligent sampling estimators for the purpose of inference, additional assumptions are required for the sake of deriving their asymptotic distribution. First, we consider a methodological issue: if the error terms around a change point are arbitrarily heteroscedastic (e.g., every error term has a distinct marginal distribution), then it is essentially impossible to estimate the distributions of the noise terms, which play a critical role in determining the asymptotic distributions of the change point estimators. Hence, we require a condition where the distribution around the change points are stable. In the sequel we assume that the error terms on either side of a change-point are i.i.d in slowly growing neighborhoods, though the distribution of errors need not be identical over the entirety of the observed data sequence. We split our results into two parts: J *fixed* and J growing with N, as the results we establish in these two cases are somewhat differently formulated.

J fixed with N: In this case, we assume the following further conditions on the model.

(M5): The jump sizes $\Delta_j := \nu_j - \nu_{j-1}$ for $j = 1, ..., J$ are also constants not dependent on N.

(M6): For every $0 \leq j \leq J$, the random variables $\{\varepsilon_{\tau_j+1}, \ldots, \varepsilon_{\tau_{j+1}}\}$ all have the same distribution as the random variable $\mathcal{E}_j$, where $\mathcal{E}_0, \mathcal{E}_2, \ldots, \mathcal{E}_J$ are fixed random variables with distributions not changing with N.

Under these two additional conditions, it is possible to characterize the asymptotic distribution of the $\hat{\tau}_j^{(2)}$'s. Prior to stating the result, we introduce the following distance function $\lambda_2(\cdot,\cdot) = \lambda_{2,N}(\cdot,\cdot)$, which accounts for the fact that at step (ISM5), the first stage subsample points are left out in the second subsample (and thus no first subsample point can be the final estimator):

$$\lambda_2(a,b) := \begin{cases} \sum_{i=1}^{N} 1\left(a < i \leq b\right) \cdot 1(i \neq k\lfloor N/N_1 \rfloor \text{ for any integer } k) & \text{if } a \leq b \\ -\sum_{i=1}^{N} 1\left(b < i \leq a\right) \cdot 1(i \neq k\lfloor N/N_1 \rfloor \text{ for any integer } k) & \text{otherwise} \end{cases}. \tag{17}$$

In terms of this distance function, $\hat{\tau}_j^{(2)}$'s converge as follows:

THEOREM 4. *Suppose conditions (M1) to (M6), and the consistency condition (14) are satisfied. Define the independent random variables L_j for $1 \leq j \leq J$ as*

$$\begin{aligned} L_j^* &:= \arg\min_{k \in \mathbb{Z}} Z_j(k) \\ Z_j(k) &:= \begin{cases} |\Delta_j|(\varepsilon_{j,1}^* + \cdots + \varepsilon_{j,k}^*) + k\Delta_j^2/2, & k > 0 \\ 0, & k = 0 \\ -|\Delta_j|(\varepsilon_{j-1,k+1}^* + \cdots + \varepsilon_{j-1,0}^*) + k\Delta_j^2/2, & k < 0\,, \end{cases} \end{aligned} \tag{18}$$

where each $\{\varepsilon_{j,k}^* : k \in \mathbb{Z}\}$ is a set of iid random variables with the same distribution as the random variable $\mathcal{E}_j$, for $j = 0, \ldots, J$.

The deviations $\left\{\lambda_2\left(\tau_j, \hat{\tau}_j^{(2)}\right)\right\}_{j=1}^{\hat{J}}$ jointly converge to the distribution of $(L_1^*, ..., L_J^*)$. That is, for any integers $k_1, \ldots, k_J$,

$$\mathbb{P}\left[\hat{J} = J,\, \lambda_2\left(\tau_j, \hat{\tau}_j^{(2)}\right) = k_j \text{ for } 1 \leq j \leq J\right] \to \prod_{j=1}^{J} \mathbb{P}\left[L_j^* = k_j\right]. \tag{19}$$

PROOF. See Section 9.8 of Supplement Part B. □

In practical terms the result enables statistical inference - construction of confidence regions.

Note that we require a finite number of change points, which is a strong restriction from a

methodological and applications point of view. Nevertheless, the result in Theorem 4 is still useful, as shown through numerical experiments in Section 5. Next, we establish a similar result for growing number of change points J.

J grows with N and Gaussian Errors: If we restrict ourselves to the case where the error terms are independent Gaussian random variables, the distribution of the intelligent sampling estimators can be characterized *even if* $J \to \infty$. Specifically, assume:

(M4-Gaussian): $\varepsilon_1, \ldots, \varepsilon_N$ are independent zero mean Gaussian random variables, with variances bounded below by $\sigma^2_{\min}$ and above by $\sigma^2_{\max}$, where $\sigma_{\min}$ and $\sigma_{\max}$ are positive constants not dependent on N.

Further, also consider conditions (M1) to (M3), and (M5). Note that (M5) along with this new condition implies that we can write $\mathcal{E}_j \sim N(0, \sigma_j^2)$ for some constants σ_js, lying between the values $\sigma_{\min}$ and $\sigma_{\max}$, a convention we will use for the remainder of the section.

When the number of parameters and (the corresponding) estimates go to ∞ as N increases, there is no fixed distribution to converge to, hence to characterize the large sample distribution, our result use probability bounds and the quantiles of a growing number of distributions. For all $\alpha \in (0,1)$ and positive values $\Delta, \sigma_1, \sigma_2$, let $Q_{\Delta,\sigma_1,\sigma_2}(1-\alpha)$ be the $1-\alpha$ quantile of $|L_{\Delta,\sigma_1,\sigma_2}|$, where

$$
\begin{aligned}
L_{\Delta,\sigma_1,\sigma_2} &= \arg\min_{t\in\mathbb{Z}} X_{\Delta,\sigma_1,\sigma_2}(t) \\
X_{\Delta,\sigma_1,\sigma_2}(t) &= \begin{cases} t\frac{|\Delta|}{2} + \sum_{i=1}^{t} \varepsilon_i^* & t > 0 \\ 0 & t = 0 \\ \frac{|t\Delta|}{2} + \sum_{i=0}^{t-1} \varepsilon_{-i}^* & t < 0 \end{cases} \\
&\text{where } \varepsilon_i^* \overset{\text{iid}}{\sim} N(0,\sigma_1^2) \text{ for } i \le 0 \text{ and } \varepsilon_i^* \overset{\text{iid}}{\sim} N(0,\sigma_2^2) \text{ for } i > 0\,. \qquad (20)
\end{aligned}
$$

One might envisage a result of the form:

$$
\mathbb{P}\left[\hat{J} = J;\ \left|\lambda_2\left(\tau_j, \hat{\tau}_j^{(2)}\right)\right| \le Q_{\Delta_j,\sigma_{j-1},\sigma_j}(\sqrt[J]{1-\alpha}) \text{ for all } j = 1,\ldots,J\right] \to 1-\alpha, \qquad (21)
$$

where $\lambda_2 := \lambda_{2,N}$ was defined in (17). However, as the distribution of $\arg\min_{t\in\mathbb{Z}} X_{\Delta,\sigma_1,\sigma_2}(t)$ is discrete, it is generally not possible to get the probabilities exactly equal to $\sqrt[J]{1-\alpha}$. Instead, we derive the following result:

THEOREM 5. *For any $\alpha \in (0,1)$ and positive values $\Delta, \sigma_1, \sigma_2$, define:*

$$
P_{\Delta,\sigma_1,\sigma_2}(1-\alpha) = \mathbb{P}\left[\left|\arg\min_{t\in\mathbb{Z}} X_{\Delta,\sigma_1,\sigma_2}(t)\right| \le Q_{\Delta,\sigma_1,\sigma_2}(1-\alpha)\right]. \qquad (22)
$$

Suppose that conditions (M1) to (M3) are satisfied, conditions (M4-Gaussian) and (M6) are

satisfied, and the first stage estimates satisfy (14) with a ρ_N such that $J\rho_N \to 0$. Then

$$\begin{aligned} &\mathbb{P}\left[\hat{J} = J;\ \left|\lambda_2\left(\tau_j, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{|\Delta_j|-2\rho_N,\sigma_{j-1},\sigma_j}(\sqrt[J]{1-\alpha}) \text{ for all } j = 1\ldots, J\right] \\ &= \left(\prod_{j=1}^{J} P_{|\Delta_j|-2\rho_N,\sigma_{j-1},\sigma_j}(\sqrt[J]{1-\alpha})\right) + o(1)\,, \end{aligned} \tag{23}$$

and

$$\begin{aligned} &\mathbb{P}\left[\hat{J} = J;\ \left|\lambda_2\left(\tau_j, \hat{\tau}_j^{(2)}\right)\right| \leq Q_{|\Delta_j|+2\rho_N,\sigma_{j-1},\sigma_j}(\sqrt[J]{1-\alpha}) \text{ for all } j = 1\ldots, J\right] \\ &= \left(\prod_{j=1}^{J} P_{|\Delta_j|+2\rho_N,\sigma_{j-1},\sigma_j}(\sqrt[J]{1-\alpha})\right) + o(1)\,. \end{aligned} \tag{24}$$

Proof. See Section 9.7 for a long and detailed proof of Supplement Part B. A shorter proof sketch which highlights the key steps is also provided in Section 9.6. □

Remark 5. *The proof of this result relies on probability bounds for the $L_{\Delta,\sigma_1,\sigma_2}$ distributions, specifically bounds on their tail probabilities, as well as bounds on the probability that $L_{\Delta,\sigma_1,\sigma_2} \neq L_{\Delta',\sigma_1',\sigma_2'}$ when $(\Delta,\sigma_1,\sigma_2) \neq (\Delta',\sigma_1',\sigma_2')$. Using the Gaussianity assumption, such probability bounds can be methodically derived. The additional condition requiring $J\rho_N \to 0$ stems from the details of the proof, but this is satisfied by several existing change point methods, including binary segmentation that will be showcased later on in the paper.*

4. Practical Implementation of Intelligent Sampling. In the previous section, we laid out a generic scheme for intelligent sampling which requires the use of a multiple change point estimation procedure on a sparse subsample of the *data-sequence*. Recall that any procedure that satisfies (13) can be used here. A variety of such procedures have been explored by various authors (see, e.g., [22], [9], [7], and [2]), and therefore a number of options are available. For the sake of concreteness, we pursue intelligent sampling with seeded binary segmentation with narrowest over threshold selection (henceforth abbreviated to "SBS") employed at Step (ISM3). This procedure was introduced in [15], with one of its main advantage being its computational scaling at an optimal rate of $O(N^* \log(N^*))$ when applied to a data sequence of length N^*. We later discuss other potential options.

Before we begin, we remind the reader that this section deals with the first stage subsample and not the whole data set. We will henceforth use the $*$ notation in connection with the quantities involved at the first stage. We let $\nu_j^* := \nu_j$ for $j = 0, \ldots, J$ and $\rho_{N^*}^* := \rho_N$, and referring back to notation used in step (ISM2) and (ISM3), we consider the sub-dataset $Z_1, \ldots, Z_{N^*}$ as a multiple change point model, with change points τ_j^*'s and levels ν_j^*'s, complying with conditions (M1) to (M4) for all large N (as a consequence of $Y_1, \ldots, Y_N$ satisfying conditions (M1) to (M4)). The minimum distance between consecutive change points would be $\delta_{N^*}^*$,which by the calculation in

(12), is greater than $K(N^*)^{1-\Xi/\gamma}$ for some positive constant K. Using this notation, (13) translates to the requirement that a change point estimation scheme applied upon $Z_1, \ldots, Z_{N^*}$ procures estimates $\hat{\tau}_j^*$'s and $\hat{\nu}_j^*$'s (which can be used as $\hat{\nu}_j^{(1)}$'s in (ISM3)) such that

$$\mathbb{P}\left[\hat{J} = J,\ \max_{j=1,\ldots,J}\left|\hat{\tau}_j^* - \tau_j^*\right| \le w^*(N^*),\ \max_{j=0,\ldots,J}\left|\hat{\nu}_j^* - \nu_j^*\right| \le \rho_{N^*}^*\right] \to 1, \tag{25}$$

for some sequences $w^*(N^*)$ and $\rho_{N^*}^*$ such that $w^*(N^*) \to \infty$, $w^*(N^*) = o(\delta_{N^*}^*)$, and $\rho_{N^*}^* \to 0$ as $N^* \to \infty$.

4.1. *A Brief Description of SBS.* Consider the model given in (11). For any positive integers $1 \le s \le b < e \le N^*$, let $n = e - s + 1$ and define the Cumulative Sum (CUSUM) statistic at b with endpoints $(s, e]$, $s < e$ being positive integers not more than N^*, as

$$\bar{Z}_{(s,e]}(b) = \sqrt{\frac{e-b}{n(b-s+1)}}\sum_{t=s+1}^{b} Z_t - \sqrt{\frac{b-s+1}{n(e-b)}}\sum_{t=b+1}^{e} Z_t.$$

These CUSUM statistics are maximized over a number of intervals, so we denote the maximizer over any interval $(s, e]$ as

$$\hat{b}_{(s,e]} := \underset{b\in\{s+1,\ldots,e-1\}}{\arg\max}\left|\bar{Z}_{(s,e]}(b)\right| \tag{26}$$

More specifically, they will be maximized over the seeded intervals, which are

$$\begin{aligned} \mathcal{I} &= \bigcup_{k=1}^{\lceil \log_{1/a}(N^*)\rceil} \mathcal{I}_k \qquad \text{where} \\ \mathcal{I}_k &= \bigcup_{i=1}^{n_k}\left\{\left(\lfloor (i-1)s_k\rfloor, \lceil (i-1)s_k + l_k\rceil\right]\right\}, \end{aligned} \tag{27}$$

where $n_k = 2\lceil a^{-(k-1)}\rceil - 1, l_k = N^* a^{k-1}$, and $s_k = (N^* - l_k)/(n_k - 1)$ for some fixed constant $a \in [1/2, 1)$ (the authors of [15] recommended $a = \sqrt{1/2}$ in practice, which we will assume from this point). SBS is performed with the procedures:

1. Fix a threshold value ζ_{N^*} and initialize the estimate set $\hat{\underline{\tau}}^* = \emptyset$ and the seeded intervals $\mathcal{I}$ as defined in (27).
2. Find the maximum and argmax of the CUSUM statistics for each $(s, e] \in \mathcal{I}$.
3. Take every interval which has that a maximum CUSUM greater than ζ_{N^*}, i.e.

$$SS := \left\{(s, e] \in \mathcal{I} : \bar{Z}_{(s,e]}(\hat{b}_{(s,e]}) \ge \zeta_{N^*}\right\}. \tag{28}$$

Take an interval of shortest length within this set longer than 2, a $(s', e'] \in SS$ where $2 \le e' - s' \le e - s$ for any other $(s, e] \in SS$ where $e - s \ge 2$, and add $\hat{b}_{(s',e']}$ to the list of estimates by setting $\hat{\underline{\tau}}^* \leftarrow \hat{\underline{\tau}}^* \cup \{\hat{b}_{(s',e']}\}$.

4. Eliminate from $\mathcal{I}$ any other intervals which contains the interval arrived at the previous step; set

$$\mathcal{I} \leftarrow \mathcal{I} - \big\{(s,e] \in \mathcal{I} : (s',e'] \subseteq (s,e]\big\} \tag{29}$$

5. Repeat steps 3 and 4 until every interval in $\mathcal{I}$ has a maximal CUSUM value less than ζ_{N^*} or length less than 2, in other words $\bar{Z}_{(s,e]}(\hat{b}_{(s,e]}) < \zeta_{N^*}$ or $e - s < 2$ for every $(s,e] \in \mathcal{I}$.

In order for SBS to yield consistent estimators, it requires the following additional condition:

(M4 (SBS)) The error terms of the data sequence are i.i.d. $N(0,\sigma^2)$, where σ is a positive constant not dependent on N.

In other words, we're restricting ourselves to iid Gaussian error terms.

LEMMA 1. *Suppose that conditions (M1) to (M3) and (M4 (SBS)) are satisfied. There exist constants C_1 and C_2 such that if SBS is performed at step ISM3 on the subsample $Z_1,\dots,Z_{N^*}$ with $\zeta_{N^*} = C_1\sqrt{\log(N^*)}$, then*

$$\mathbb{P}\Big[\hat{J} = J;\quad \max_{i=1,\dots,J} |\hat{\tau}_i^* - \tau_i^*| \le C_2 \log(N^*)\Big] \ge 1 - C_3/N^*\,. \tag{30}$$

for some constant C_3. The computational complexity of this SBS procedure will be $O(N^*\log(N^*))$.

REMARK 6. *The above result is adapted from Theorems 2 and 3 of [15] which applies to the more general setting where $\underline{\Delta}$, the minimum signal jump, can decrease to 0 as $N \to \infty$.*

This result gives us consistent change point estimates, but it falls short of the condition we imposed at (25), which requires consistent signal estimates as well. Fortunately it is possible to obtain consistent signal estimators by simply averaging between the change point estimates,

$$\hat{\nu}_j^* = \frac{1}{\hat{\tau}_{j+1}^* - \hat{\tau}_j^*}\left(\sum_{\hat{\tau}_j^* < i \le \hat{\tau}_{j+1}^*} Z_i\right) \qquad \text{for } j = 0,...,\hat{J} \tag{31}$$

with the convention of $\hat{\tau}_0^* := 0$ and $\hat{\tau}_{\hat{J}+1}^* := N^*$. Obtaining all $\hat{J}$ of the signal estimators has $O(N^*)$ computational complexity, and these estimators are consistent:

LEMMA 2. *Suppose conditions (M1) to (M3) and (M4 (SBS)) are satisfied. There exist constants C_1 and C_2 such that if SBS is performed at step ISM3 on the subsample $Z_1,\dots,Z_{N^*}$ with $\zeta_{N^*} = C_1\sqrt{\log(N^*)}$, and signal estimates are obtained as in (31), then there exists a sequence $\rho_{N^*}^* \to 0$ such that*

$$\mathbb{P}\left[\hat{J} = J;\quad \max_{i=1,\dots,J} |\hat{\tau}_i^* - \tau_i^*| \le C_2\log(N^*);\quad \max_{i=0,\dots,J} |\hat{\nu}_i^* - \nu_i^*| \le \rho_{N^*}^*\right] \to 1 \tag{32}$$

as $N^* \to \infty$. We also know that $C_2\log(N^*) \to \infty$ and $C_2\log(N^*) = o(\delta_{N^*}^*)$, as[4] $\delta_{N^*}^* > K(N^*)^{1-\Xi/\gamma}$ for some constant $K > 0$, and $1 - \Xi/\gamma > 0$, thus satisfying the conditions of (13).

[4]By the calculation of (12) and also stated earlier in Section 4.

PROOF. See Section 9.10 in Supplement Part B. □

REMARK 7. *The condition that $J\rho_N \to 0$, required by Theorem 5, can also be satisfied by imposing further conditions than (M1)-(M3) and (M4 (SBS)), namely condition (M7 (SBS)) found in Section 9.1. Specifically, the value of Ξ from condition (M3) has to be restricted. For additional details, refer to Section 9.10 of the supplementary materials.*

The consistency result of Lemma 2 matches our requirement, but it presents difficulties in practice. Specifically, while the constant C_2 is known to exist, its actual value is unspecified. One way to resolve this issue, at least for asymptotic considerations, would be to use intervals slightly greater in order than $\log(N^*)$. The results of the Lemma implies that

$$\mathbb{P}\left[\hat{J} = J; \quad \max_{i=1,\ldots,J} |\hat{\tau}_i^* - \tau_i^*| \leq \beta_{N^*} \log(N^*); \quad \max_{i=0,\ldots,J} |\hat{\nu}_i^* - \nu_i^*| \leq \rho_{N^*}^*\right] \to 1, \tag{33}$$

for any sequence $\beta_{N^*} \to \infty$, and we can simultaneously satisfy $\beta_{N^*} \log(N^*) \to \infty$ and $\beta_{N^*} \log(N^*) = o(\delta_{N^*}^*)$ if we set β_{N^*} to grow slowly[5]. One possible choice would be to set $\beta_{N^*} = \log(N^*)$.

REMARK 8. *We discuss a more precise and methodologically sound way to construct intervals after step ISM3 in Section 9.1.*

4.2. *Computational Considerations.* The order of computational time is more complicated than in the single change point case, owing to the fact that the procedure can involve a growing number of data intervals at the second stage. For the sake of our analysis, we make the simplifying assumption that $\delta_N/N^{1-\Xi} \to C_1$ and $J(N)/N^{\Lambda} \to C_2$ for some $\Lambda \in [0, \Xi]$ and some positive constants C_1, C_2.

Under the previous assumptions, we consider performing our two stage procedure, with the first stage using SBS to procure estimates in step (ISM3), and then the second stage of the procedure following through steps (ISM5) and (ISM6). We state here (full details in Section 9.2 of the supplement) that for any fixed Ξ and Λ, the total computational time of these two stages will be the larger of $O\left(N^{\frac{1+\Lambda}{2}} \log^2 N\right)$ or $O\left(N^{\Xi+\eta} \log N\right)$, where η can be set to any small value between 0 and $1 - \Xi$. We note that in the special case where $\Lambda = 0$ and $\Xi = 0$, which translates to finitely many ($J \nrightarrow \infty$) change points that are roughly spaced evenly apart, we would have the total computational be $O(\sqrt{N} \log^2 N)$. This time scaling would be the shortest time order for the two stage procedure.

To achieve lower computational time order, we can also consider multiple (> 2) stage procedures. A three stage procedure would have the same first stage (SBS or some other procedure on an even subsample); the second stage estimate would estimate a stump model within each confidence interval from stage 1, but using a subsample from each interval instead of the whole interval of

[5]More specifically, so that $\log \beta_{N^*} = o(\log N^*)$.

data, to create another set of confidence intervals; at stage 3 stump models would be fitted with all the points within each confidence interval created in stage 2. A four and five stage procedure can be similarly designed. In terms of running time, a k-stage procedure can, in principle, run in almost $O(\sqrt[k]{N}\log N)$ time in certain cases, but for methodological concerns and for illustrative purposes we focused on the two stage procedure. Further details and discussions regarding more stages can be found in Section 9.3 of the Supplement.

5. Performance Evaluation of Intelligent Sampling: Simulation Results. We next present a series of simulation results. Part of our computational work illustrates the validity of our theoretical work, such as the rate of convergence, the lower than $O(N\log(N))$ computational running time, and the asymptotic distributions. We also present some other simulations to analyze the practical benefits of our procedure. All simulations in this section were performed on a server with 24 Intel Xeon ES-2620 2.00 GHz CPUs, with a total RAM of 64 GB.

5.1. *Rate of Convergence and Time.* Our first showcased set of simulation aims to confirm the predicted scaling of the maximal deviation (no faster than $\log(J)$) and computational time ($O(\sqrt{N}\log(N))$ in the best cases). We simulate a sequence of data sequences with increasing N, and to show that the $\sqrt{N}\log(N)$ time scaling is reachable, we set the number of change points to be growing slowly and the separation between consecutive change points to be somewhat close to being equal.

We generated sequences of length N varying from 10^5 to $10^{7.5}$, evenly on the log scale, with the number of change points being $J \approx \log_{10}(N)^2$. The change point location and the signal levels were randomly generated:

- The spacings $(\tau_1, \tau_2-\tau_1, \tau_3-\tau_2, ..., N-\tau_J)$ were generated as the sum of the constant $\frac{N}{1.5J}$ and the rounded values of $\left(N-\frac{(J+1)N}{1.5J}\right)\cdot(U_{(1)}, U_{(2)}-U_{(1)}, \ldots, U_{(J)}-U_{(J-1)}, 1-U_{(J)})$ where $U_{(1)}, \ldots, U_{(J)}$ are the ordered variables from J iid uniform $[0,1]$ variables.
- The signals were generated as a Markov chain with ν_0 initialized as 0, and iteratively, given ν_i, ν_{i+1} is generated from a density proportional to $f(x) = \exp(-0.3(x-\nu_i-\underline{\Delta}))1(\nu_i+\underline{\Delta} \le x \le M) + \exp(0.3(x+\nu_i-\underline{\Delta}))1(\nu_i-\underline{\Delta} \ge x \ge -M)$, where M was taken to be 10 and $\underline{\Delta}$ was taken to be 1.

For each of 10 values of N, 50 configurations of change points and signals were generated, and on each of those configurations 40 datasets with iid $N(0,1)$ error terms simulated. At stage 1 we performed binary segmentation[6] (denoted BinSeg for short) and the steps outlined in Section 9.1, along with some ad-hoc additional steps to increase accuracy. In detail, these steps are

1. an even subsample was taken with $N_1 = 50\sqrt{N}$, and BinSeg was performed on this dataset with tuning parameter $\zeta_N = N_1^{0.2}$ (which is within the range described in Lemma 1)

[6]Procedure described in [9]. It has the same order of running time and very similar consistency results as SBS. It is used here partially for convenience and partially to show that intelligent sampling is not contingent on SBS, as other procedures used at the first stage can still yield good results.

2. remove points too close together by removing every $\hat{\tau}_i^*$ where $\hat{\tau}_i^* - \hat{\tau}_{i-1}^* \leq 15$
3. estimate for the signals and remove every $\hat{\tau}_i^*$ where the estimated signal to the left and the right have a difference less than 0.5
4. perform steps (ISM4-1) to (ISM4-4)
5. repeat steps 2 and 3
6. construct confidence intervals using the $L_{\hat{\Delta}_j,\hat{\sigma},\hat{\sigma}}$ distributions and then proceed with steps (ISM5) and (ISM6)

To gauge the rate of convergence, the maximum deviations $\max_{j=1,\ldots,J} |\lambda_2(\tau_j, \hat{\tau}_j^{(2)})|$ were recorded for instances where $\hat{J} = J$ (which occurs over 99.5% of the time for all N). We also compared the mean computational times of intelligent sampling vs the mean time using BinSeg on the whole data sequence, where the latter was averaged over 100 iterations with similarly generated data (50 configurations of randomly generated change points and signals, 2 runs each configuration). The results are depicted in Figure 1 and are in accordance with the theoretical investigations: the quantiles of $\max \lambda_2 \left(\tau_j, \hat{\tau}_j^{(2)}\right)$ scale sub-linearly with $\log(J)$ (which is $\sim \log(\log(N))$ in this setup where $J \sim \log^2(N)$) as predicted in Theorem 3, and the computational time of intelligent sampling scales in the order of $\sqrt{N}$ compared to the order N computational time of using binary segmentation on the entire dataset.

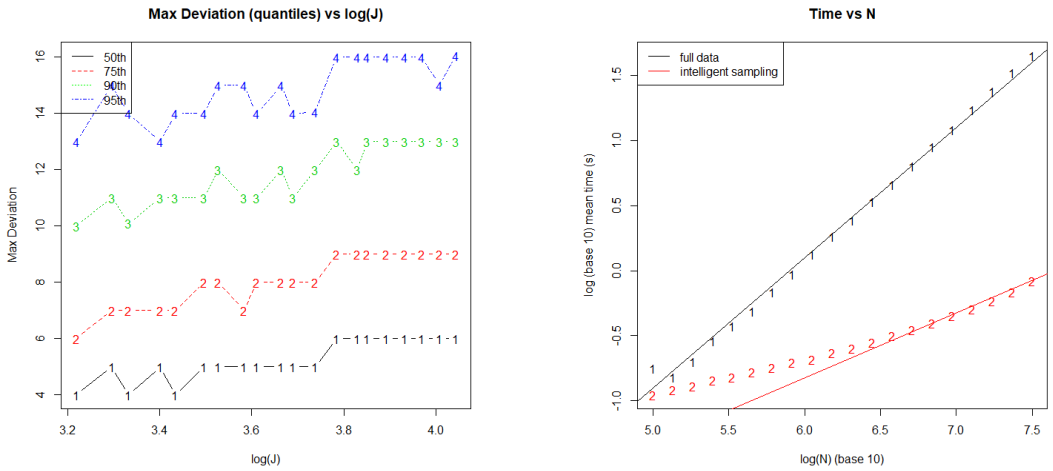


Fig 1: Left: Quantiles of the max deviations versus $\log(\log(N))$, which is the same order as $\log(J)$. Over the observed regime of parameters, the maximal deviation scales with J, as was predicted by Theorem 3. **Right:** Log-log plot of mean computational time when using intelligent sampling to obtain the final change-point estimates at stage two, and using binary segmentation on the full data to construct change-point estimates, with reference lines of slope 1 (black) and 0.5 (red) respectively. To give some sense of the actual values, for $N = 10^{7.5}$ the average time for intelligent sampling vs full data were, respectively, 0.644 and 31.805 seconds.

5.2. *Asymptotic Distribution.* A second set of simulation experiments was used to illustrate the asymptotic distribution of the change point deviations. We considered a setting with $N = 10^7$ and 55 change points, which was the maximum number of change points used in the last simulation setting. However, we consider different error term distributions and placement of change points in 4 different settings:

(Setup 1): one set of signal and change point locations generated as in the previous set of simulations, with i.i.d. $N(0,1)$ error terms

(Setup 2): change points evenly spaced apart with signals 0,1,0,1,.., repeating, and i.i.d. $N(0,1)$ error terms

(Setup 3): change points evenly spaced part with 0,1,0,1,... repeating signals, and error terms generated as $\varepsilon_i = \frac{\varepsilon_i^* + \varepsilon_{i+1}^* + \varepsilon_{i+2}^* + \varepsilon_{i+3}^* + \varepsilon_{i+4}^*}{\sqrt{5}}$ for all $i = 1, ..., N$, where the ε_i^*'s are generated as i.i.d. $N(0,1)$;

(Setup 4): change points evenly spaced with signals 0,1,0,1,..., and error terms generated from an AR(1) series with parameter 0.8, and each marginally $N(0,1)$.

The first and second setups will demonstrate the result of Theorem 4. As for the other two setups with dependent error terms, we claim that the asymptotic distribution will follow a similar form as the asymptotic distribution in the iid case, with more specific details in Section 9.4 of the supplement. For all 4 cases the first stage of intelligent sampling was performed identically as for the previous set of simulations, and with the same tuning parameters. At the second stage, first stage subsample points were omitted for data with setups 1 and 2, but not for setups 3 and 4. From 2000 iterations on each of the 4 simulation setups, the distributions of the maximum deviations (maximum of $|\lambda_2(\tau_j, \hat{\tau}_j^{(2)})|$ for the first two setups and $|\hat{\tau}_j^{(2)} - \tau_j|$ for the other two setups) are seen to match well with their predicted asymptotic distributions. To illustrate the convergence of the individual change point estimates, we also show that the distribution of the 27th change point matches with the L-type distributions appearing in Proposition 3 of the supplement.

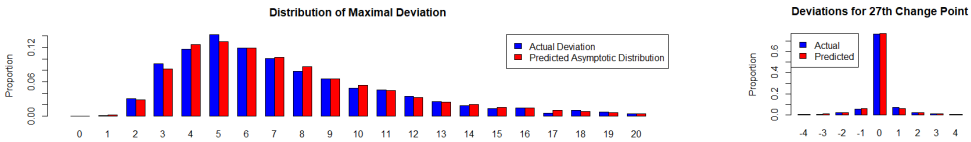


Fig 2: Distributions of $\max_{1 \le j \le 55} \lambda_2 \left(\tau_j, \hat{\tau}_j^{(2)}\right)$ (**Left**) and $\lambda_2 \left(\tau_{27}, \hat{\tau}_{27}^{(2)}\right)$ (**Right**) from simulations of setup 1.

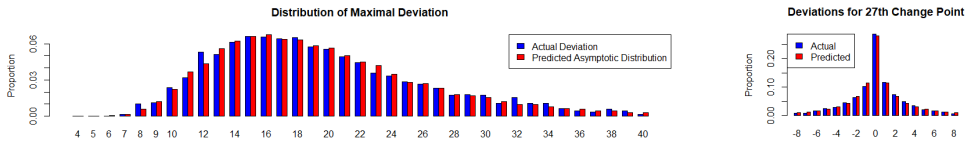


Fig 3: Distributions of $\max_{1\leq j\leq 55}\lambda_2\left(\tau_j,\hat{\tau}_j^{(2)}\right)$ (**Left**) and $\lambda_2\left(\tau_{27},\hat{\tau}_{27}^{(2)}\right)$ (**Right**) from setup 2.

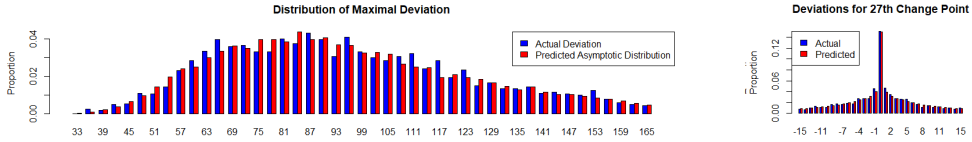


Fig 4: Distributions of $\max_{1\leq j\leq 55}\left|\tau_j-\hat{\tau}_j^{(2)}\right|$ (**Left**) and $\left|\tau_{27}-\hat{\tau}_{27}^{(2)}\right|$ (**Right**) from setup 3

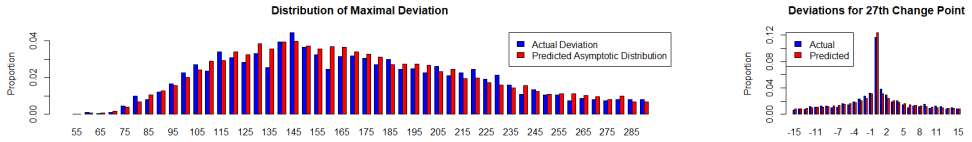


Fig 5: Distributions of $\max_{1\leq j\leq 55}\left|\tau_j-\hat{\tau}_j^{(2)}\right|$ (**Left**) and $\left|\tau_{27}-\hat{\tau}_{27}^{(2)}\right|$ (**Right**) from setup 4.

The distribution of the deviations for setup 1 is the least spread out, with the primary reason that the jump between signals is randomly generated, but lower bounded by 1, while the other 3 setups have signal jumps all fixed at 1. Setups 3 and 4 have the most spread out distributions, as the dependence among the error terms causes the estimation to be less accurate, but only up to a constant rather than an order of magnitude. Nevertheless, in all 4 setups, the change point estimates behave very closely to what Theorems 5 and 4 (for the first two scenarios) and Proposition 3 (for the third and fourth scenarios) predict

5.3. *Asymptotic Distribution for the Heteroscedastic Case.* We further explored the validity of Proposition 3 by examing a setting involving heteroscedatic errors. A data sequence of length $N=10^7$ with 55 evenly spaced changed points and signals of magnitude $0,1,0,1,0,1,\ldots,1$ was generated. Instead of generating error terms from a stationary series, we, instead, generated them as independent segments of Gaussian processes as follows. For $j=1,2,3,\ldots,$

1. from τ_{4j} to $\frac{\tau_{4j+2}+\tau_{4j+1}}{2}$ the errors are iid $N(0,1)$;

2. from $\frac{\tau_{4j+2}+\tau_{4j+1}}{2}$ to $\frac{\tau_{4j+3}+\tau_{4j+2}}{2}$ the errors are $\varepsilon_i = \frac{\varepsilon_i^*+0.5\varepsilon_{i+1}^*+0.25\varepsilon_{i+2}^*}{\sqrt{1^2+0.5^2+0.25^2}}$ where the ε_i^*'s are iid $N(0,1)$ (and will be treated as a generic iid $N(0,1)$ sequence from here on;)
3. from $\frac{\tau_{4j+3}+\tau_{4j+2}}{2}$ to τ_{4j+3}, error terms are $\varepsilon_i = 0.5 \cdot \frac{\varepsilon_i^*+\varepsilon_{i+1}^*+\varepsilon_{i+2}^*+\varepsilon_{i+3}^*}{\sqrt{4}}$;
4. from τ_{4j+3} to τ_{4j+4} the error terms are $\varepsilon_i = 0.7 \cdot \frac{\varepsilon_i^*+\varepsilon_{i+3}^*}{\sqrt{2}}$;

and the error terms generated in each stretch are independent of those in any other stretch. This creates a situation where around τ_{4j+1} the error terms are iid $N(0,1)$, around τ_{4j+2} the error terms are stationary, and around τ_{4j+3} and τ_{4j+4} the error terms are stationary to the left and to the right, but their autocorrelation and marginal variances change at the change points. With the same intelligent sampling procedure as setups 3 and 4, and the same tuning parameters, we ran 2000 replicates of this setup and recorded the $\hat{\tau}_j^{(2)} - \tau_j$ values for $j = 1, \ldots, 55$.

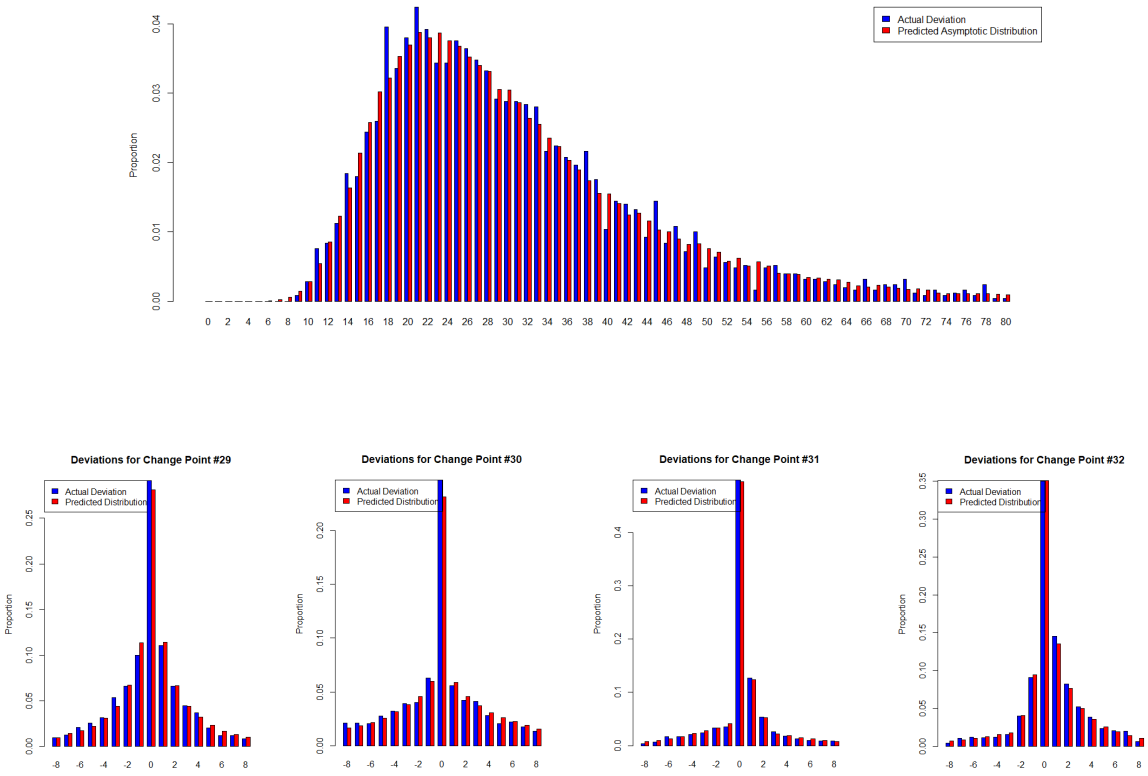


Fig 6: Top: Predicted and actual distribution of the maximal deviation $\max_{1\le j\le 55} |\hat{\tau}_j^{(2)} - \tau_j|$. **Bottom, Left to Right:** Predicted and actual distributions of the individual deviations $|\hat{\tau}_j^{(2)} - \tau_j|$ for $j = 29, 30, 31,$ and 32.

These numerical results are consistent with Proposition 3 and even for change points whose error

distributions are different to the left and the right of the corresponding change point, the deviations match up very closely with the stated asymptotic distributions.

5.4. *Choice of Subsample Size.* In order to showcase the veracity of our asymptotic results, for the previous simulation settings we tuned the first stage subsample size to ensure that the exact number of change points was detected with high accuracy. In practice, it is difficult to know how to choose the best size of the first stage subsample, since one does not know the number of change points or how far apart they are. In this section we propose a systematic *data-driven* way of determining a correct subsample size at the first stage: draw a (small) subsample of size N_1, perform BinSeg, record the number of change-points detected; then perform BinSeg on a fresh subsample whose size is enhanced by a constant factor and re-record the number of change-points. Keep increasing the subsample size and re-estimating until some stopping criterion related to the growth in the number of detected change-points is reached. Formally, we propose the following procedure, coined **"BinSeg-Double"**:

1. perform BinSeg on an even subsample with $N_1 = 2\sqrt{N}$, obtaining an estimate $\hat{J}_1$ for J along with change point estimates;
2. now generate a new subsample of size N_1 that is double the previous value and obtain an estimate $\hat{J}_2$ along with change point estimates;
3. continue the previous step until we reach a $\hat{J}_i$ such that $\hat{J}_i > \hat{J}_{i-2} + 5$ and $|\hat{J}_i - \hat{J}_{i-1}| < 5$, or until $\left|\min\{\hat{J}_{i-3}, \ldots, \hat{J}_i\} - \max\{\hat{J}_{i-3}, \ldots, \hat{J}_i\}\right| < 5$.

With regard to the stopping criteria at the last step, we reason that generally, the number of change points detected will go up at every iteration until nearly all change points are detected, or the subsample size has increased to the size of the full data set. Hence, we stop doubling the subsample once the number of detected change points essentially stabilizes. This stopping criterion has good results in several settings as we shall see in subsequent simulations, but other scenarios might require alternative stopping rules for better accuracy. In practical settings the practitioner may use a different last step to adapt to the difficulty of the problem.

We gauged the effectiveness of this method as follows. We generated data sequences of length $N = 10^6$ and iid $N(0, 1)$ error terms, with evenly spaced change points, and the signals alternating between $0, \Delta, 0, \Delta, \ldots$, where Δ is the SNR parameter, allowed to vary aross simulation settings in addition to the number of change-points. For each configuration, we invoked intelligent sampling at stage 1 using the BinSeg-Double procedure followed by curtailing consecutive change point estimates that are within 5 points of each other. As an added detail, for the sake of methodology, we used the R changepoint package functions in this and all subsequent sections, instead of the manually coded BinSeg algorithm with pre-specified tuning parameter as in the previous sections. The second stage calculations remain unaltered. We recorded the computational time and confidence level coverage for each simulation configuration, and compared them with the computational time of running BinSeg on the entire dataset.

TABLE 1
Two Stage Running Time

SNR \ J	50	100	150	200	250	300	350	400	450
1	1.67	4.6	11.84	15.1	39.58	49.8	53.95	80.41	132.22
1.5	1.08	2.85	9.04	7.42	20.12	20.84	27.34	39.3	74.16
2	0.71	1.68	3.34	7.63	11.7	20.37	24.48	45.79	57.06

Average running time in seconds across 200 iterations of every configuration.

TABLE 2
Full Data Running Time

SNR \ J	50	100	150	200	250	300	350	400	450
1	17.63	26.66	35.77	45.14	54.56	64.2	73.77	83.72	93.47
1.5	17.39	26.55	35.92	44.59	53.75	63.34	72.75	82.71	92.4
2	17.13	26.01	35.17	44.44	53.63	63.39	72.61	82.37	91.86

Average running time in seconds across 50 iterations of every configuration.

As is evident from these tables, the running time of intelligent sampling relative to the full data procedure is lowest when the number of change points is low and the SNR at these changes are high. Conversely, when the number of change points is large and/or the SNR values are low, it becomes necessary to take a bigger subsample at the first stage, increasing the relative computation time.

We also show that our proposed first stage procedure leads to accurate results. In each iteration, after the second stage change point estimates were obtained and the timer stopped, we constructed $(1-\alpha)$ level (not simultaneous) confidence intervals around each estimate, and recorded the proportion of true change points captured in the union of these intervals. Specifically, at each iteration k we constructed a confidence region $\hat{C}_k$, and recorded

$$\frac{1}{J \cdot (\#\text{ of iterations})} \sum_{k=1}^{(\#\text{ of iterations})} \sum_{j=1}^{J} 1(\tau_j \in \hat{C}_k). \tag{34}$$

TABLE 3
Change point Coverage for Two Stages

SNR \ J	50	100	150	200	250	300	350	400	450
1	0.966	0.971	0.945	0.944	0.968	0.966	0.935	0.967	0.969
1.5	0.978	0.985	0.958	0.958	0.983	0.981	0.954	0.982	0.982
2	0.993	0.997	0.972	0.971	0.996	0.995	0.971	0.995	0.996

Proportion of change points covered over all 200 iterations with $\alpha = 0.01$. As all proportions are close to 0.99, this indicates accurate estimates.

We next perform a similar set of simulations, with N and the error structure staying the same but with an added element of randomness from the placement of the change points and values of

the signals. The change points $(\tau_1, \ldots, \tau_J)$ are generated as $N/(4(J+1))$ plus the random variables $\frac{3N}{4(J+1)}(U_{(1)}, U_{(2)} - U_{(1)}, U_{(3)} - U_{(2)}, \ldots, 1 - U_{(J)})$, where $U_{(1)}, \ldots, U_{(J)}$ are the ordered variables from J iid uniform (0,1) random variables. The changes in signal at the different change point are randomly generated as independent $(2S-1) * V$ variables, where S is Bernoulli with $p = 0.5$ and V is uniform on [1,4].

TABLE 4
Running Time for Random Design

Method \ J	50	100	150	200	250	300	350	400	450
Two Stage	1.02	3.3	9.84	21.67	32.06	44.7	64.09	86.89	109.84
Full Data	18.07	27.36	36.81	46.38	56.04	65.77	75.64	85.32	95.16

Average running time in seconds across 200 iterations of the random design setup.

The coverage proportion of actual change points are

TABLE 5
Coverage Proportion for Random Design

J	50	100	150	200	250	300	350	400	450
Coverage	0.985	0.986	0.993	0.994	0.995	0.996	0.996	0.995	0.996

Proportion of change points covered by $\alpha = 0.01$ level (non-simultaneous) confidence sets across 200 iterations.

We note that in both simulation setups, intelligent sampling gains the most computational time savings over the full data analysis when the separation between change points is long (corresponding to smaller J) and when the SNR is high. Such setups require only a small subsample at stage 1 to capture all the change points. In certain cases where J is high enough and/or the SNR low enough, *it would be impossible to detect all the change points at stage 1 with any strictly smaller subsample, meaning BinSeg-Double would continue until the full dataset is utilized.* In this respect, the doubling procedure is *automatically calibrated* to the intrinsic difficulty of the problem.

Our findings are perfectly consistent with our previous claims that intelligent sampling is most useful for detecting sparsely placed, high SNR change points in long data sequences; in settings with densely placed, lower SNR change points (or a large number of change points), one can not obtain accurate results without utilizing large subsamples (or even the entire data set), thus mitigating, or even nullifying, the computational time savings of intelligent sampling.

6. Data Applications. We apply the intelligent sampling framework to two network traffic data sets. In the first one that exhibits a stationary behavior, we artificially inject change points, by shifting the local mean of the data across different stretches. Hence, there is a known ground truth to calibrate the performance of intelligent sampling in a setting where other features of the data (error distributions, presence of temporal dependence) may not fully adhere to the assumptions used in establishing the theoretical properties of the procedure. The second data set is a fully observed data set where the ground truths are unknown.

6.1. *Partly Emulated Data.* The effectiveness of the proposed intelligent sampling procedure is illustrated on an Internet traffic data set, obtained from the public CAIDA repository `http://data.caida.org/datasets/passive/passive-oc48/20020814-160000.UTC/pcap/` that contains traffic traces from an OC48 (2.5 Gbits/sec) capacity link. The trace under consideration contains traffic for a two hour period from a large west coast Internet service provider back in 2002. The original trace contains all packets that went through the link in an approximately 2 hour interval, but after some aggregation into bins of length 300 microseconds, the resulting data sequence comprises $N = 1.5 \times 10^7$ observations. After applying a square-root transformation, a small segment of this sequence is depicted in Figure 7 and some of its statistical characteristics in Figure 8, respectively.

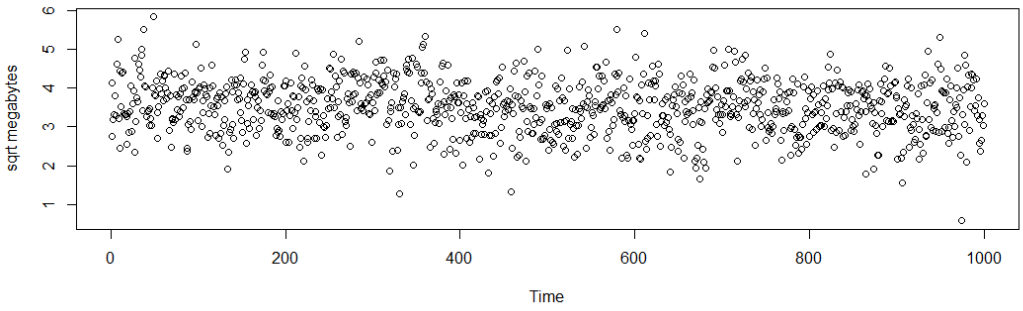


Fig 7: First 5000 time points of the data after a square root transformation.

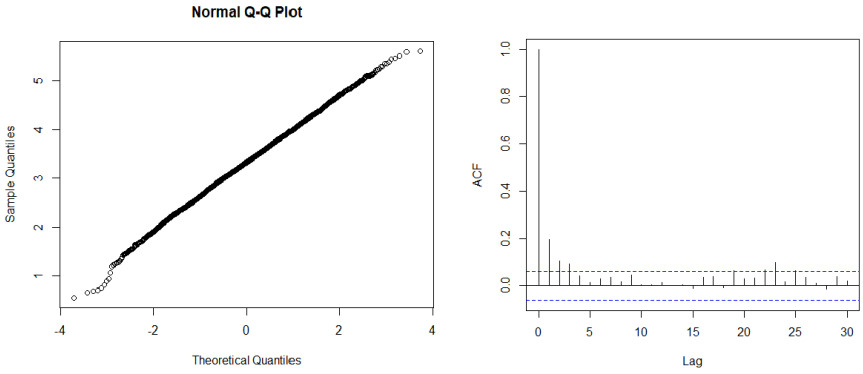


Fig 8: QQ plot and estimated ACF of first 5000 points of data set after square root transformation. Note the normality of the transformed data.

It can be seen that the data are close to marginally normally distributed, while their autocorrelation

decays rapidly and essentially disappears after a lag of 10. Similar exploratory analyses performed for multiple stretches of the data lead to similar conclusions. Hence, for the remainder of the analysis, we work with the square-root transformed data and model them as a short range dependent sequence.

To illustrate the methodology, we used an *emulation* setting, where we injected various mean shifts to the mean level of the data of random durations, as described next. This allows us to test the proposed intelligent sampling procedure, while at the same time retaining all features in the original data.

In our emulation experiments, we posit that there are two types of disruptions, short term spikes that may be the result of specific events (release of a software upgrade, a new product or a highly anticipated broadcast) and longer duration disruptions that may be the result of malicious activity [5, 10]. We first take the post square root transformation data $Y_1, \ldots, Y_N$ and add long, persistent change points by taking $Y_i \leftarrow Y_i + \theta_i$, where θ_i is piecewise constant with change points $(\tau_1, \ldots, \tau_J)$ $(J = 200)$ generated as $\frac{N}{2.5(J+1)} + \frac{1.5N}{2.5(J+1)} \cdot (U_{(1)}, U_{(2)}, \ldots, U_{(J)})$, where $(U_{(1)}, \ldots, U_{(J)})$ are the order statistics of J iid uniform (0,1) variables, and signals $(0, S_1, 0, S_2, \ldots)$ where $S_1, S_2, \ldots$ are iid uniform (0.7,2) variables. Next we emulate a larger number of *spikes* in the data, or short term large increases in the signal, by randomly selecting 400 locations for these spikes as $(V_1, \ldots, V_{400}) = (N - 50) \cdot (U_1, \ldots, U_{500})$ (where $U_1, U_2, \ldots$ are iid uniform (0,1)), and setting $(Y_{V_j}, \ldots, Y_{V_j+50}) \leftarrow (Y_{V_j}, \ldots, Y_{V_j+50}) + W_j$ for all j, where W_j is an iid uniform (10,15) variable. A depiction of a segment of the data with the emulated signal is given in Figure 9.

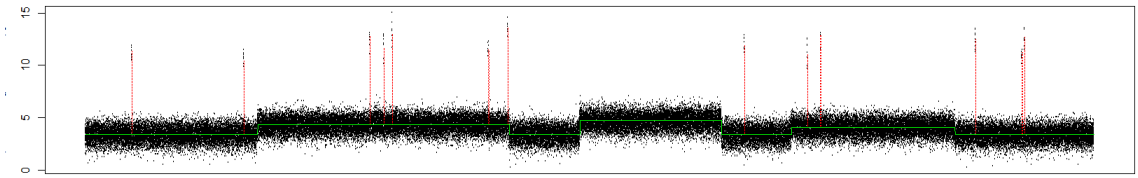


Fig 9: Example of the first half million points of an emulated data example.

As mentioned in the introduction, the main objective of the proposed methodology is to identify *long duration, persistent shifts* in the data sequence using a limited number of data points; in the emulation scenario used, this corresponds to change points (τ_1, τ_J), while we remain indifferent to the points $V_j, V_j + 50$ for $j = 1, \ldots, 400$, which as previously described are change points corresponding to the spikes[7]

The two-stage intelligent sampling procedure was implemented as follows: (i) the first stage used BinSeg-Double , followed by removal of estimates less than 4 points apart, followed by steps (ISM4-1) to steps (ISM4-4), and then followed by removal of estimates less than 16 apart. (ii) For

[7]We note that the theoretical development does not include spiky signals. Nonetheless, we included spiky signals in our emulation to mimic the pattern of internet traffic data. As will be seen later, our method is quite robust to the presence of this added feature.

each j, the second stage interval surrounding $\hat{\tau}_j^{(1)}$ was chosen to have half width $Q_{\hat{\Delta}_j,\hat{\sigma},\hat{\sigma}}\left(1-\frac{0.01}{\hat{J}}\right)$ where $\tilde{\Delta}_j$'s and $\hat{\sigma}$ are estimates of the jump and the common standard deviation. A stump model was then fitted to the data in each second stage interval to obtain the final estimates of the change-points and the final (2nd stage) CIs were constructed.

To assess the accuracy, we calculated the coverage proportion of the 90%, 95%, and 98% level confidence intervals over different emulation settings. To construct these confidence intervals we had to randomly generate data sequences with identical distribution structure as the data (which would give us a random sample of L-type distributions and their quantiles). We generated these sequences as marginally normal random variables, with marginal standard deviation the same as the sample sd of the first 50,000 points of the 1.5×10^7 length data. Finally, the ACF of the generated series was matched with the sample ACF of the first 50,000 points up to a lag of 20: we first generated vectors of iid normal variables, then multiplied them with the Cholesky square root of the band matrix created with the sample ACF (the bandwidth of this matrix is 20, and non-zero entries are taken from the first 20 values of the sample ACF).

Intelligent sampling exhibits highly satisfactory performance: among all 200 change points corresponding to persistent changes, the average coverage probability for the 90%, 95%, and 98% nominal confidence intervals were 0.874, 0.914, and 0.939 respectively. On the other hand, for change points induced by the spikes, the average coverage probability was lower than 0.035 even for the 98th confidence interval. However, since the focus of intelligent sampling is on long duration persistent signals, missing the spiky signals is of no great consequence. In terms of computational burden, the average emulation setting utilized 17.1% of the full dataset, requiring an average time of 29.7 seconds to perform the estimation.

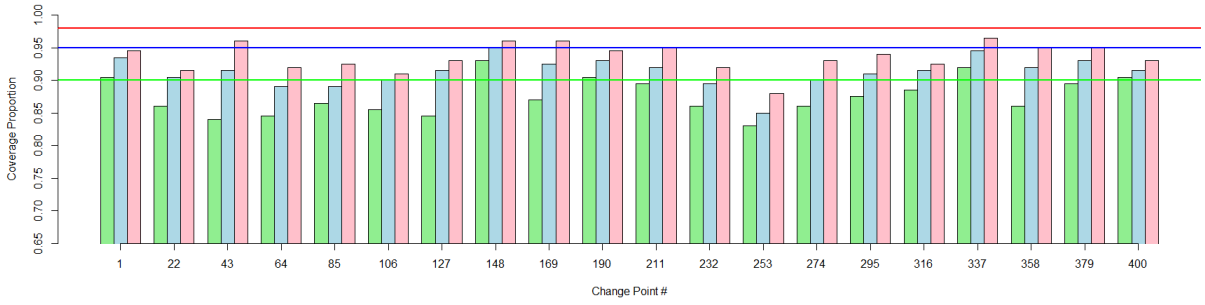


Fig 10: Coverage proportions, the proportion of time when the change point was covered by some confidence interval, for the 90% level (green bars), 95% level (blue bars), and 98% level (red bars) within the 500 iterations, for a select number of 20 change points (change point # 2 is always the second one in order, # 3 is the third in order, etc). Horizontal reference lines are at 0.9 (green), 0.95 (blue), and 0.98 (red).

6.2. *Real Data Example.* We further gauged the practical performance of intelligent sampling by evaluating its performance when applied to a different data set with *naturally occurring change points* of varying durations. The growing capacity of Internet links (routinely at 10 Gbps) together with proliferation of such links have rendered network monitoring a more challenging but at the same time critical task, given increased malicious activities. However, most monitoring tools rely on sampling packets at rates of 1:1000 or 1:10000 in order to minimize interference with network infrastructure. In an experimental project, an alternative that aggregates traffic at 2 sec intervals and then records it was developed at Merit Network (see details on the technology in [10]). The data analyzed next correspond to a trace of all incoming traffic to the autonomous system administered by Merit Network and corresponds to a data sequence obtained during the spring of 2019, comprising of 2080768 observations. The objective is to identify persistent change points in the traffic pattern. However, since the data have not been previously analyzed, no ground truth is known.

Unlike the 2002 trace previously analyzed, the current trace (segments in logarithmic scale shown in Figure 11) exhibits a richer set of patterns, but broadly resembles a piecewise linear signal plus noise, with multiple discontinuities and abrupt changes in slopes.

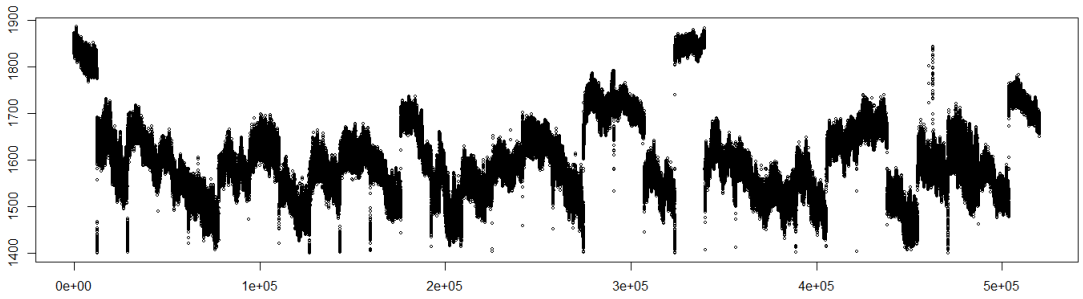


Fig 11: Plot of the last quarter of the data (i.e., points $0.75N$ to N of the data).

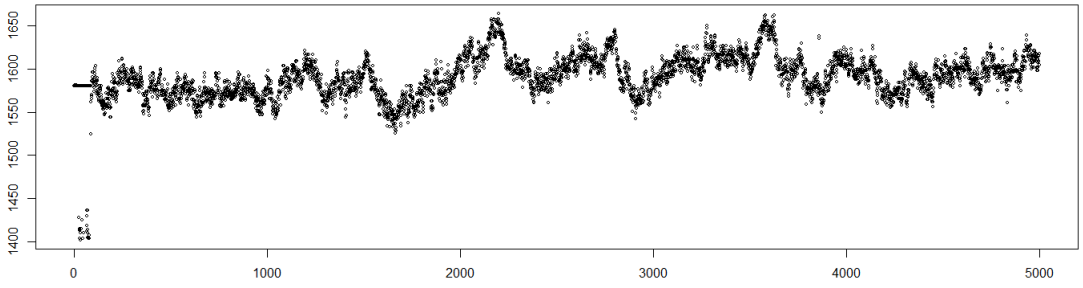


Fig 12: Plot of the first 5000 points of the data.

The data set contains several particularly prominent change points exhibiting very large jumps, but zooming in to short intervals reveals that the signal oscillates up and down with a wavelength

of a few hundred points, as shown in Figure 12. Therefore, the data also contain numerous less prominent change points in signal slope at the crest and troughs of the signal vibrations. For our analysis, we are more interested in the locations where large jumps are located. Detection of *all minor signal changes* requires a thorough analysis of the entire data set.

Analysis of this data set via intelligent sampling is conceptually similar to data sets with piecewise constant stretches that have been the primary focus of this paper. The multiple change point algorithm in Stage 1 must now be able to deal with piecewise linear signals. Once the change-points from Stage 1 are localized, a piecewise linear function with a single jump can be fitted at Stage 2 in each local neighborhood to get updated estimates of the change points and corresponding confidence intervals by simulating from an estimated error distribution. Specifically, instead of the BinSeg-Double procedure described in Section 5.4 in the first stage, we use a doubling procedure that performs Narrowest-Over-Threshold (denoted as NOT henceforth) detection (introduced in [3]) after each doubling. The NOT procedure functions similarly to BinSeg. At each step, a CUSUM-like expression is maximized over sub-intervals of the data[8]. A change point is declared if the maximum exceeds a specified threshold value, after which the segment is bisected by the new change point estimate and the process repeats over the two components. Because we are trying to detect only the biggest changes in signals instead of all of them, we can use the value of the threshold tuning parameter to limit the number of change points detected by the algorithm. In the associated `R` package `not`, we performed this tuning by varying the `q.max` parameter of the `features` function, but we set all other tuning parameters to their default values. The stopping criteria for the first stage was slightly altered from before.[9]

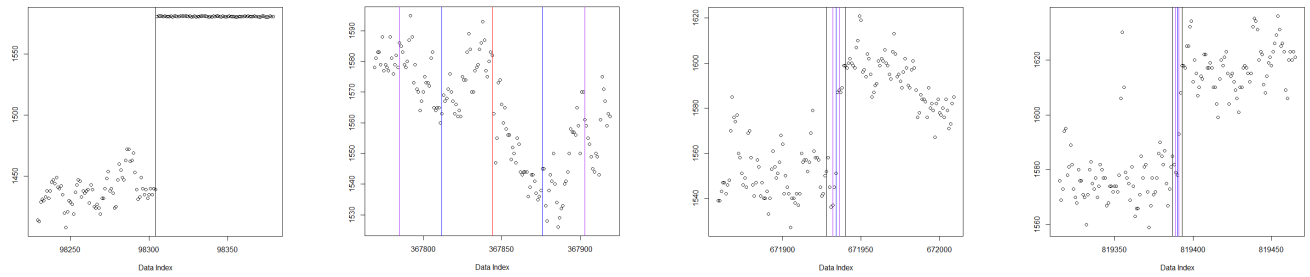


Fig 13: A few selected estimates (red) and confidence intervals (blue denotes 90% level, purple for 95%, and black for 99%). To construct the confidence levels, for each change point estimate we fit 2 AR time series models using 75 points on the left and 75 points on the right.

We are interested in comparing the performance of intelligent sampling with NOT versus applying NOT over the whole data. Thus, over several values of the `q.max` tuning parameter, we performed estimation 40 times each for intelligent sampling and full data estimation. All computing was

[8] To be very specific, the maximization is performed over random intervals contained in the segment.

[9] Another stopping criterion was added: if the number of change points ever decreases, stop and keep the previous results.

performed on a desktop with an Intel Core i7-8700K CPU, with a script that utilized the `parallel` package in `R`. We remind the reader that because of the randomized nature of the NOT algorithm, results can vary between the different runs of estimation.

TABLE 6
Average running time over 40 iterations.

Method \ q.max	50	75	100	125	150	175	200
Two Stage	12.03	4.81	8.57	9.94	21.88	8.52	11.73
Full Data	463.3	475.94	494.75	517.46	503.08	500.86	507.12

TABLE 7
Average number of change points detected over 40 iterations.

Method \ q.max	50	75	100	125	150	175	200
Two Stage	49.8	72.8	77.3	80.83	81.2	77.55	80.42
Full Data	49.85	70.12	76.33	78.75	83.1	80.62	79.92

In terms of the average number of change points detected, intelligent sampling and the full data analysis detected around the same number of change points for every scenario. When `q.max` is low, both methods detected close to the maximum number of change points allowed by the package function, but as is clear from Table 6, the number of detected change points stabilized at around 80 for both procedures, irrespective of the value of `q.max`. Table 7 gives a snapshot of the time comparison of the full data analysis with intelligent sampling, with the latter generally seen to be between 40 to 60 times faster.

To see whether these detected change points correspond to the larger change points, we also show the distribution of the estimated jump sizes.

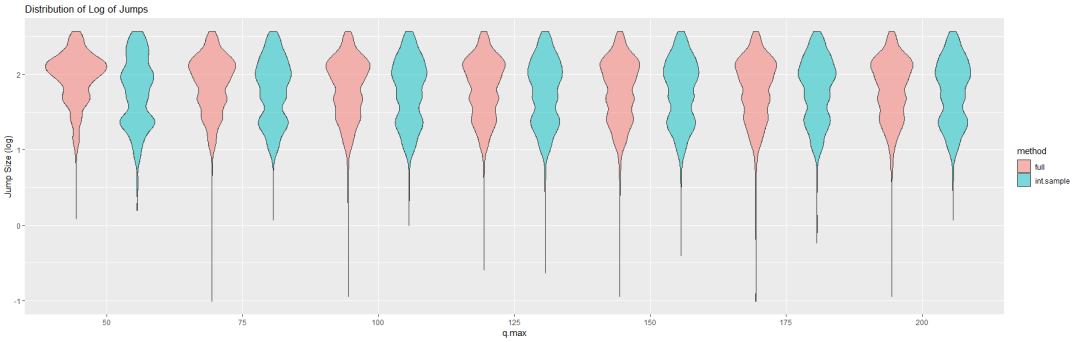


Fig 14: Distribution of the log (base 10) of the absolute values of jump sizes.

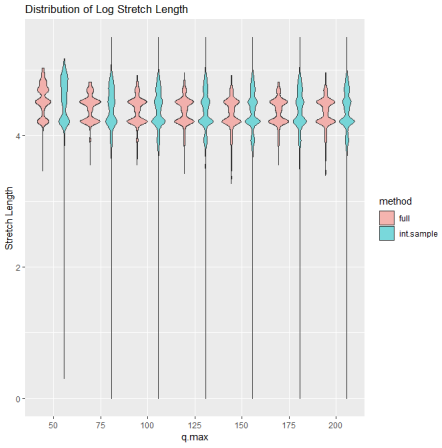


Fig 15: Distribution of the log (base 10) of the distances between consecutive change point estimates.

TABLE 8
Proportion of Large Jumps

q.max	Two Stage	Full Data
50	0.55	0.78
75	0.59	0.67
100	0.57	0.62
125	0.56	0.61
150	0.57	0.58
175	0.6	0.61
200	0.57	0.61

Proportion of jumps that are greater than 50. The threshold of 50 was chosen because the data in Figure 11 seems to oscillate with an amplitude of no more than 50 when far from the change points.

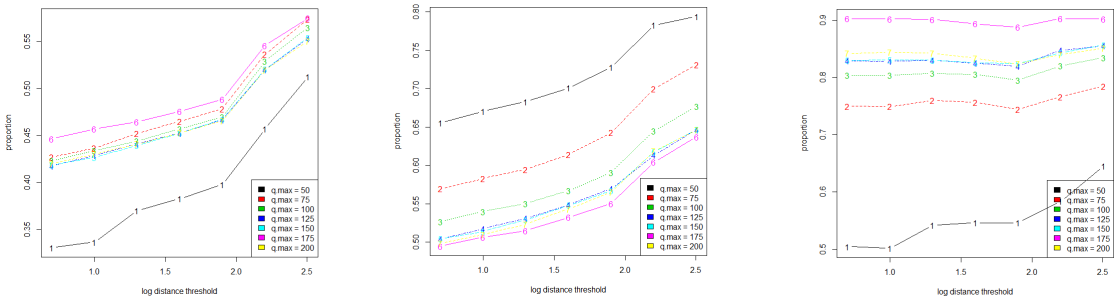


Fig 16: Left: Proportion of estimates from intelligent sampling that are close to the estimates from full data estimation, calculated using the expression in (35). **Center:** A comparison between half of the full data estimates versus the other half, using an expression similar to (35). **Right:** The ratio of the values in the left graph divided by the values in the center graph.

For each `q.max` value, the distribution of the logarithm of the jump sizes are similar, with the jump distribution from the intelligent sampling procedure being skewed more towards the lower values. The proportion of detected jumps which exceed 50 (a value that appears to be a reasonable upper bound on the noise amplitude from inspection of Figure 14) is also similar between the two estimation methods. It is clear that most of the detected jumps are large. Figure 15 demonstrates the distributions of distances between consecutive change points, and it is clear from the plots that the vast majority of inter change point distances obtained by intelligent sampling are between 10000 and 10000 points, which shows that our approach is picking up long and persistent changes, exactly what it is designed for.

Another way of comparing the change point estimates procured from intelligent sampling is to quantify the proportion of intelligent sampling estimates that are close to some full data estimate. For each fixed `q.max` value, we have the full data estimates $\{\hat{\tau}_{i,f}^{(j)} : i = 1, \ldots, \hat{J}_{j,f}\}$ and the intelligent sampling estimates $\{\hat{\tau}_{i,s}^{(j)} : i = 1, \ldots, \hat{J}_{j,s}\}$ for iterations $j = 1, \ldots, 40$. We fix a bandwidth parameter $\delta \in \{5, 10, 20, \ldots, 320\}$, and calculate the following proportions to gauge how similar the two sets of estimates are:

$$\frac{1}{40}\sum_{k=1}^{40}\left[\frac{1}{\sum_{j=1}^{40}\hat{J}_{j,s}}\sum_{\substack{j=1,\ldots,40\\ i=1,\ldots,\hat{J}_{j,s}}} 1\left(\min_{\ell=1,\ldots,\hat{J}_{k,f}}\left|\hat{\tau}_{\ell,f}^{(k)} - \hat{\tau}_{i,s}^{(j)}\right| < \delta\right)\right]. \tag{35}$$

These values can be seen in the left panel of Figure 16 for a variety of parameter values, and they come fairly close to 60%. Although this proportion is well under 100%, this is primarily due to the randomness of performing NOT estimation on this specific data set. Indeed, if we split the full data estimates with the first 20 iterations in one group and the last 20 iterations in the other group, then comparison of the two groups of the full data estimates with an expression similar to (35) will still result in values not exceeding 60%. Taking the ratios of the proportions for each `q.max` and δ values, we see that for large `q.max` values the ratios are very close to 1. In other words, the comparison of intelligent sampling estimates to the full data estimates show no more dis-similarity than when comparing the full data estimates across disjoint runs.

For drawing downstream conclusions from an analysis of this type, one should next flag the change-point locations that are persistent across the different estimation runs, and subject them to a more refined local analysis. We however do not proceed in that direction, primarily because the point of this analysis is somewhat different – namely, the effectiveness of intelligent sampling compared to full data analysis on real data.

In conclusion, the above analysis amply demonstrates that intelligent sampling was able to extract a similar number of change points as a full data analysis on the entire data, including finding a large proportion of estimators which also shows up in a full data estimate. It was able to do this while using much less time, as the running time data demonstrates.

7. Concluding Remarks and Discussion. This paper introduced sampling methodology that reduces significantly the computational requirements in multi-change point problems, while not compromising on the statistical accuracy of the resulting estimates. It leverages the *locality principle*, which is obviously at work in the context of the classical signal-plus-noise model employed throughout this study. Intelligent sampling is devised specifically for detecting major but relatively infrequent changes in a data stream, what one can think of as significant 'regime changes', and should *not* be relied upon to identify short-lived disturbances or small perturbations to the mean level of a data stream. While the paper has dealt with a one-dimensional data sequence, extensions to problems involving multiple (potentially high-dimensional) data [e.g., sequences produced by

cyber-physical systems equipped with a multitude of sensors monitoring physical or man-made phenomena] are of obvious interest. Also, while our theoretical development studies the canonical model with piecewise flat stretches between change-points, extensions to allow variations in the mean level between change-points, e.g. piecewise polynomial or nonparametric specifications of the mean function, while requiring more careful estimation of mean levels, pose no added conceptual difficulties: indeed, our second data set uses a piecewise linear model. Finally, the focus in this paper has primarily been on a two-stage procedure, which is easiest to implement in practice and suitable for many applications. Nevertheless, as illustrated in Section 9.3, in specific settings involving data sets of length exceeding 10^{10} points, a multi-stage procedure may be advantageous.

A key technical requirement for intelligent sampling is that the procedure used to obtain the 1st stage estimates needs to exhibit consistency properties, e.g. (14). The choice of utilizing SBS and BinSeg, is due to their computational attractiveness and the fact that they readily provide consistent estimates of the number of change points and their locations. Yet other procedures which exhibit similar consistency results and run in $O(N \log N)$ time would be just as useful.

We now briefly turn our attention to two other popular multiple change point methods and their potential relevance to the intelligent sampling problem. Two popular for models defined as in (11) are the estimation of multiple structural breakpoints introduced in [2] and PELT as described in [12]. The method described in [2] does give consistent estimates, but only under the much stricter condition that J is a constant and there exists values $\beta_1, \ldots, \beta_J \in (0,1)$ such that $\tau_j = [\beta_j N]$ for all $j = 1, \ldots, J$ and N. Further, to run the actual procedure would require the use of dynamic programming which is computationally expensive ($O(N^2)$ time). With the PELT procedure, the implementation itself runs in a more manageable $O(N)$ time; however, this works under the very different Bayesian setting where the spacings $\tau_{j+1} - \tau_j$ are iid generated from some distribution. Further, PELT was built upon a procedure described in [23], which examines another Bayesian model where every point $\{1, \ldots, N\}$ has a probability p of being a change point, and the development did not go into details regarding rates of convergence of the change point estimates. Due to the theoretical and computational restrictions of the multiple structural breakpoints method and the differing framework under which PELT works, we focused our analysis on binary segmentation.

We also mention the SMUCE procedure, introduced in [7], where lower probability bounds for the events $\mathbb{P}[\hat{J} \neq J]$ and $\mathbb{P}[\max_{j=1,\ldots,J} \min_{i=1,\ldots,\hat{J}} |\hat{\tau}_i - \tau_j| \leq c_N]$, for any sequence c_N, were derived. These results can be combined to yield $\mathbb{P}[\hat{J} = J; \max_{j=1,\ldots,J} |\hat{\tau}_j - \tau_j| \leq c_N] \to 1$ under certain restrictions and for some sequences c_N that are $o(\delta_N)$, and therefore could be used in the first stage of intelligent sampling. SMUCE has the flexibility of working for a broader class of error terms [10] but as was stated in [7], the procedure involves dynamic programming which runs in $O(N^2)$ time. This last point is less of an issue for a modified version of SMUCE designed for iid Gaussian errors with heterogeneous variances. H-SMUCE, in [18], could run the procedure in as low as $O(N)$ time in some cases. Overall, SMUCE could be used as the first part of intelligent sampling, and the regimes of δ_N and restrictions on the subsample size N_1 needed for intelligent sampling to be consistent could be fleshed out in a similar manner as in this paper. However, as BinSeg and WBinSeg are

[10] Some results apply when the errors are iid from a general exponential family.

somewhat easier to implement computationally, we chose to perform our analysis with them instead.

Our simulation results indicate that for non-Gaussian errors, as well as dependent error terms (with or without Gaussian distributions), the deviations of our estimators behave like the L-type distributions, even in J growing with N settings. This suggests that our results could be extended to broader classes of error terms, and future work could consider models incorporating errors with dependence structures and/or with non-Gaussian distributions. Extending Theorem 5 to these settings would require an in-depth investigation into probability bounds on the argmin of a drifted random walk with non-i.i.d and/or non-Gaussian random components along the lines of the derivations in Section 10 of the Supplement for random walks with Gaussian innovations. We speculate that this work would be more amenable to rigorous analysis when the tail probabilities of the error terms decay exponentially (similar to Gaussian distributions) and when the dependence is local , e.g. m-dependence, where each error term is only correlated with the m neighbors to its left and right.

In conclusion, any procedure used at stage 1 of intelligent sampling puts restrictions on the model specifications, as consistent second stage estimators cannot be obtained if the first stage procedure is not consistent. Established results for SBS and BinSeg, as in [15], [22], and [9], consider only the i.i.d Gaussian framework. Extending these approaches to a more flexible class of error terms therefore requires theoretical exploration of these procedures properties beyond Gaussian errors, or using alternative methods at stage 1 which do not need the Gaussian error framework, e.g. [2] and [7].

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